

The cycle type of two rows of a random Latin square: exact identities and a reduction of the weak Cavenagh–Greenhill–Wanless conjecture

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Abstract

Let L be a uniformly random Latin square of order n and σ the permutation of columns carrying its first row to its second. Cavenagh, Greenhill and Wanless conjectured that the cycle type of σ is asymptotically that of a uniform derangement, in total variation. We prove three exact identities and a reduction theorem. The marked-pair identity expresses the ratio of completion counts of adjacent cycle types as the odds of one bit (“rows 1, 2 lie in a common column cycle of a marked column pair”) in a single uniform square; a reduction theorem, by which the conjecture follows from that bit being fair to $o(1/\log n)$; and the ladder identity, by which the bit is exactly fair except on the squares in which no “ladder pair” of rows is flippable. An orbit method bounds the probability of the exceptional set by a product of conditionally independent factors. The conjecture is thereby reduced to a “witness lemma” (Hypothesis 5.1): that with probability $1 - o(1/\log n)$ some ladder pair of a long ladder has a short row cycle through one marked column avoiding the other — a lower-tail statement about rare local configurations at rows selected by the square, whose heuristic expectation is $\gtrsim \log^2 n$. The orbit method gives alternative sufficient conditions of the same type, one of which trades the rarity for a genericity constant; and at the first ladder pair alone it gives a third exact identity, expressing the bias $\mathbb{P}_X[B] - \mathbb{P}_X[A]$ as an expectation over the event that none of the column pairs along the row permutation of that pair admits a legal toggling turn, so that the conjecture also follows if that event has probability $o(1/\log n)$ (in samples at $n \leq 100$ it is $\approx 5.5/n$, which says nothing about the required rate). We state these precisely, report the numerical evidence (up to $n = 100$, including, through a sampler of completions of a fixed rectangle, types of exponentially small probability), and locate the remaining difficulty within the known switching technology: every exact identity we have found inside the conditioned space is fair, and the lower bounds needed are of the kind whose proof in [1] has a gap. Along the way we observe that the published proof of the upper bound in Lemma 3.12 of Cavenagh–Greenhill–Wanless has a gap for every split other than $(2, 2)$; our results do not depend on that bound, and we re-derive from the surviving direction the tail estimate on the number of cycles that the reduction theorem needs.

1 Introduction

Let P_n be the law of the cycle type of $\sigma = \sigma_{1,2}(L)$ for a uniform Latin square L of order n , and Q_n the law of the cycle type of a uniform derangement of $[n]$. Cavenagh, Greenhill and Wanless [1, Conj. 6.1] conjectured:

Conjecture 1.1. $d_{\text{TV}}(P_n, Q_n) = o(1)$ as $n \rightarrow \infty$.

For $\lambda \vdash n$ with all parts ≥ 2 let $C_n(\lambda)$ be the number of completions of a fixed $2 \times n$ Latin rectangle of cycle type λ to a Latin square, divided by $(n-2)!$ (the count does not depend on the rectangle chosen, since rectangles of the same type are isotopic by a row-preserving isotopy), and let $\gamma(\lambda)$ be the number of permutations of type λ . Then $P_n(\lambda) \propto \gamma(\lambda)C_n(\lambda)$ and $Q_n(\lambda) \propto \gamma(\lambda)$, so the conjecture says that C_n is nearly constant on the types that carry most of the mass. CGW proved $C_n(\lambda)/C_n(\mu) \in [\frac{1}{2}, \frac{3}{2}]$ for adjacent types (one part of λ split into two parts of μ , both ≥ 2), and used it to show that the two laws agree within a factor $n^{3/2}$ on every type with at most $\frac{6}{5} \log n$ parts; the upper bound $\frac{3}{2}$, and with it the upper half of the latter statement, is affected by the gap described in Remark 3.4 and [2].

Status of the arguments. The proofs below have been checked by automated referees (independent language-model agents with access to the sources and the data) and, where feasible, by exhaustive computation at $n \leq 7$ and by sampling (this includes §6(e), added last); no human referee has read them.

Results. Throughout, a *mark* is a column pair $P = \{j, j'\}$ with $j' = \sigma^\alpha(j)$ on an m -cycle of σ , $m = \alpha + \beta$, $\alpha, \beta \geq 2$; $X = X(n, \lambda; \alpha, \beta)$ is the set of marked squares of type λ ; the *frame* of P is the permutation of rows $\pi(r) = r'$ with $L(r', j') = L(r, j)$; and P is a *B-pair* if rows 1, 2 lie in one cycle of π , an *A-pair* otherwise. Our results are:

- (1) *Marked-pair identity* (Theorem 2.4): $2C_n(\lambda)/C_n(\mu) - 1 = \mathbb{P}_X[B]/\mathbb{P}_X[A]$ exactly, for every $n, \lambda, \alpha, \beta$.
- (2) *Reduction* (Theorem 3.3): if $|\mathbb{P}_X[B]/\mathbb{P}_X[A] - 1| \leq \delta(n)$ for all adjacent pairs with $\alpha \neq \beta$ (equivalently $|\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{1}{4}\delta(1 + o(1))$), then $d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)})$. The tail estimate it needs on the number of cycles of σ is proved from the trivial half of (1) alone (Proposition 3.2).
- (3) *Ladder identity* (Theorem 4.2): with the ladder pairs $(x_k, y_k) = (\pi^{-k}(1), \pi^{-k}(2))$, $1 \leq k < K_0$, and “flippable” meaning that j, j' lie in different cycles of the permutation ρ_{x_k, y_k} of columns induced by rows x_k, y_k ,

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset], \quad |\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{1}{2}\mathbb{P}_X[\text{Flip} = \emptyset].$$

So Conjecture 1.1 follows from (L): $\mathbb{P}_X[\text{Flip} = \emptyset] = o(1/\log n)$.

- (4) *Witness lemma* (Proposition 5.2): (L), hence the conjecture, follows from Hypothesis 5.1: with probability $1 - o(1/\log n)$, a long ladder contains a pair (x_k, y_k) whose ρ_{x_k, y_k} -cycle through j has length $\leq \log^5 n$ and avoids j' (such a pair is flippable outright), and, for the offset diagonals of short-arc instances, that a constant fraction of the expected number of diagonals carry one. This is the reference hypothesis of the paper.
- (5) *Orbit method* (Propositions 6.3 and 6.6): the flippabilities of a family of ladder pairs are conditionally independent given the orbit of a group of commuting column-cycle turns. With a fixed matching of columns as coordinates this gives $\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X \exp(-\frac{1}{2} \sum_{k \in I} \bar{y}_k)$ with an explicit “orbit weight” $\bar{y}_k$; with the *adaptive* coordinates $(\rho_k^t(j), \rho_k^t(j'))$ — the t -th columns along the permutation $\rho_k = \rho_{x_k, y_k}$ of the k -th ladder pair from the marked columns — every coordinate is separated by the mark, so its turn toggles whenever it is legal and clean, and $\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X[2^{-|U|}]$, where U is the set of ladder pairs having

a short clean candidate cycle (subject to a collision condition). These give two alternative sufficient conditions (Hypotheses 6.4 and 6.7), for which we see no implication to or from Hypothesis 5.1: the first needs a genericity constant that the switchings available inside X cannot provide, the second needs $C \log \log n$ rare local events on a long ladder where the witness lemma needs one.

- (6) *The first ladder pair alone* (Propositions 6.10 and 6.11): at the first pair (x_1, y_1) the adaptive turn needs no side conditions, and combined with the first ladder trade it gives

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{E}_X[(\mathbf{1}_B - \mathbf{1}_A)(\mathbf{1}_{F^c} - \mathbf{1}_F); G^c], \quad |\mathbb{P}_X[B] - \tfrac{1}{2}| \leq \tfrac{1}{2}\mathbb{P}_X[G^c],$$

where F is the flippability of (x_1, y_1) and G the event that one of the $\nu - 1$ column pairs $\{\rho^t(j), \rho^t(j')\}$ along $\rho = \rho_{x_1, y_1}$ (ν = the length of the shorter arc or cycle of ρ at the mark) induces a permutation of rows that separates x_1 from y_1 with a legal cycle. So the conjecture also follows from Hypothesis 6.12: $\sup_{\lambda, \alpha, \beta} \mathbb{P}_X[G^c] = o(1/\log n)$ — an event selected by one row pair, with candidates of constant probability (heuristically $\Theta(n)$ of them; in samples of the type (n) at $n = 30, 50, 100$, $\mathbb{P}_X[G^c] \approx 5.5/n$) in place of a long ladder of rare events. Again no implication to or from Hypothesis 5.1 is known.

All four hypotheses are of one logical type: a lower bound on the probability of a configuration at rows or columns selected by the frame of the mark, inside the space conditioned on the type of rows 1, 2. Their heuristic margins are large; none of the exact tools of this paper gives a lower bound of that kind (§7). The data (Jacobson–Matthews samples, and a sampler of completions of a fixed $2 \times n$ rectangle that reaches types of P_n -probability $e^{-\Theta(n)}$, Remark 6.14) give $\mathbb{P}_X[B] = \frac{1}{2} \pm 0.003$ in every tested class at $n \leq 50$ and witness events that are independent to three digits along the ladder; they do not reach the regime of any of the hypotheses, and the empirical sizes quoted for the hypothesised quantities are first-moment information only.

A gap in CGW’s Lemma 3.12. Case 3 of the splitting procedure in [1] (cross-switch at $\{\omega j, \omega j'\}$, then backflip at $\{j, j'\}$) does not behave as claimed: the cross-switch reverses the ω -arc from ωj to $\omega j'$ (or the complementary arc, depending on the smallest corner symbol), which contains j' (resp. j), so $\{j, j'\}$ ends at distance 2. For $\min(\alpha, \beta) \geq 3$ the backflip then lands in the wrong type; for $\min(\alpha, \beta) = 2 < \max(\alpha, \beta)$ the type is right but the edge is not an edge of the joining multigraph, so the identity $G_S = G_J$ on which the double count rests fails (already at $n = 5$, where we computed both multigraphs exactly). We give the details in a separate note [2]; here we only record (Remark 3.4) which published statements depend on it. Nothing in the present paper does: the lower bound $\frac{1}{2}$ of CGW’s lemma is the trivial half of Theorem 2.4, and the upper bound $\frac{2}{3}$ is, by the same theorem, $\mathbb{P}_X[B] \leq \frac{2}{3}$ — a weak form of what we are after. By Theorem 4.2 it would follow from $\mathbb{P}_X[\text{Flip} = \emptyset] \leq \frac{1}{3}$.

Conventions. $L(r, c)$ is the entry in row r , column c . For rows x, y , $\rho_{x, y}$ is the permutation of columns with $L(y, \rho_{x, y}(q)) = L(x, q)$, so that $\sigma = \rho_{1, 2}$ up to inversion; a *row cycle* of (x, y) is a cycle of $\rho_{x, y}$. For columns q, c , $\delta_{q, c}$ is the permutation of rows with $L(\delta(r), c) = L(r, q)$, and its cycles are the *column cycles* of (q, c) . A *trade* of rows x, y along a row cycle swaps their entries on the columns of that cycle; a *turn* of a column cycle swaps the entries of columns q, c in its rows. Both are Latin trades (the result is again a Latin square). For $\lambda \vdash n$, $S(\lambda)$ is the set of Latin squares of order n with first row the identity whose rows 1, 2 have cycle type λ , so $|S(\lambda)| = \gamma(\lambda)(n - 2)!C_n(\lambda)$; we also write $D_\lambda = \gamma(\lambda)$ for the number of derangements of

type λ . A column pair $\{j, j'\}$ is *admissible* if $j' \notin \{j, \sigma(j), \sigma^{-1}(j)\}$ (CGW's condition for the pair operations; every marked pair is admissible since $\alpha, \beta \geq 2$). Following CGW, the *flip* at an A -pair whose columns lie in different σ -cycles, and the *backflip* (or *unflip*) at an A -pair whose columns lie in one σ -cycle, is the turn of the column cycle of the pair through row 2 (it joins, resp. splits, the σ -cycles involved); the *switch* at a pair whose columns lie in different σ -cycles is the trade of rows 1, 2 along the row cycle through one of the two columns, selected by CGW's rule (the cycle whose columns carry the smaller minimum symbol). We call Conjecture 1.1, which asks only for convergence in total variation, the *weak* conjecture; the stronger statements suggested by Cameron ("tends very rapidly") and by the data (pointwise ratios $1 + o(1)$) are not addressed here. Jacobson–Matthews sampling [3] is used for all numerical data; the scripts and logs are in the project repository.

2 The marked-pair identity

Throughout, squares are normalised to have first row id; CGW's column map ω , defined by $L(1, \omega(j)) = L(2, j)$, is then $\omega(j) = L(2, j)$, and we take $\sigma = \omega$ (only the cycle type matters, and it is the same for ω and ω^{-1}). For the proofs it is convenient to work with *unnormalised* squares (equivalently, symbol-relabelling classes): every instance count below then acquires a common factor $n!$, which cancels from every ratio, and the maps we use (flip, switch) act on unnormalised squares without any renormalisation step. Statuses, cycle types, and column positions are relabelling-invariant, so we freely state results in the normalised picture.

Definition 2.1 (marked instances). Let $\lambda \vdash n$ contain a part $m = \alpha + \beta$ ($\alpha, \beta \geq 2$), and let μ be λ with one part m split into α, β . The *splitting instance space* is

$$X = X(n, \lambda; \alpha, \beta) = \{(L', P) : L' \in S(\lambda), P = \{j, \sigma^\alpha(j)\} \text{ with } j \text{ in an } m\text{-cycle of } \sigma(L')\},$$

unordered pairs counted once. Thus every L' carries exactly $\lambda_m m$ marked pairs when $\alpha \neq \beta$ and $\lambda_m \alpha$ when $\alpha = \beta$. Since $\alpha, \beta \geq 2$ every marked pair is admissible, and $X = X_A \sqcup X_B$ according to the A/B status of P in L' . The *joining instance space* is

$$J = J(n, \mu; \alpha, \beta) = \{(L, P) : L \in S(\mu), P = \{j, j'\}, j \text{ in an } \alpha\text{-cycle}, j' \text{ in a different } \beta\text{-cycle}\},$$

again with $J = J_A \sqcup J_B$ by the status of P in L .

The counting arithmetic is immediate: with D_ν the number of derangements of cycle type ν and $N(\nu) = C_n(\nu)(n-2)!$,

$$\frac{2|X|}{|J|} = \frac{2 D_\lambda N(\lambda) \lambda_m m}{D_\mu N(\mu) \mu_\alpha \mu_\beta \alpha \beta} = \frac{2 C_n(\lambda)}{C_n(\mu)} \quad (\alpha \neq \beta), \quad (1)$$

using $D_\lambda/D_\mu = \alpha \mu_\alpha \beta \mu_\beta / (m \lambda_m)$; for $\alpha = \beta$ one has $D_\lambda/D_\mu = \alpha^2 \mu_\alpha (\mu_\alpha - 1) / (m \lambda_m)$, $|J| = |S(\mu)| \binom{\mu_\alpha}{2} \alpha^2$ and $|X| = |S(\lambda)| \lambda_m \alpha$, and (1) holds verbatim. The content of the identity is therefore the following count.

Lemma 2.2 (cycle trades preserve pair status). *Let $P = \{j, j'\}$ be an admissible column pair of a Latin square L and let C be one column cycle of P . Trading C (swapping the entries of columns j, j' in the rows of C) inverts the pair permutation π_P on C and fixes it elsewhere; the partition of rows into column cycles of P is unchanged. In particular the A/B status of P itself is invariant under the CGW flip at P , which trades the full column cycle through row 2.*

Proof. Write $\pi = \pi_P$, so $\pi(r)$ is the row of column j' holding the symbol $L(r, j)$; C is π -closed and the trade swaps $L(r, j) \leftrightarrow L(r, j')$ for $r \in C$. Fix $r \in C$ and let π' be the new pair permutation. The new $L(r, j)$ is the old $L(r, j')$; the row of the new column j' holding that symbol is found by solving (new $L(s, j')$) = (old $L(r, j')$): for $s \in C$ the new $L(s, j')$ is the old $L(s, j)$, and old $L(s, j) = \text{old } L(r, j')$ exactly when $\pi(s) = r$. Hence $\pi' = \pi^{-1}$ on C ; off C nothing changes. A cycle inverted in place has the same support, so the partition into cycles is unchanged. $\square$

Lemma 2.3 (switch toggle). *Let $(L, P) \in J$ and let Q be the row cycle through one column of P (the two row cycles are distinct), selected by CGW's min-symbol rule. Trading Q (the CGW switch) changes π_P by precomposition with the transposition of rows 1, 2: the cycles of π_P through rows 1, 2 are split if rows 1, 2 are in one cycle, and joined if in two. Hence the switch toggles the status of P , preserves $S(\mu)$ and the instance data (it inverts one σ -cycle in place), and is an involution: the trade preserves the traded cycle's symbol set (the two rows carry the same symbols on its columns), so the min-symbol rule re-selects the same cycle. Consequently $|J_A| = |J_B|$, exactly.*

Proof. Say Q is the row cycle through j ; its columns include j but not j' . The trade swaps the row-1 and row-2 entries in the columns of Q ; for π_P only column j matters, where the entries of rows 1, 2 are exchanged. Thus $\pi'_P = \pi_P \circ \tau_{12}$, and multiplying by a transposition splits or joins the cycles through its two points. Rows 1, 2 lie in one cycle of π_P exactly when P is a B -pair. The trade swaps the two rows' entries along the columns of Q , which inverts the corresponding σ -cycle in place: the cycle type and the column sets of all row cycles are preserved, so $(L, P) \mapsto (\text{sw}(L), P)$ maps J to J . $\square$

Theorem 2.4 (marked-pair identity). *For every n , λ , and split $m = \alpha + \beta$ with $\alpha, \beta \geq 2$ (including $\alpha = \beta$),*

$$|J_A| = |J_B| = |X_A|, \quad \text{hence} \quad \boxed{\frac{C_n(\lambda)}{C_n(\mu)} = \frac{|X|}{2|X_A|}, \quad 2 \frac{C_n(\lambda)}{C_n(\mu)} - 1 = \frac{|X_B|}{|X_A|}.}$$

In particular $C_n(\lambda) > \frac{1}{2} C_n(\mu)$ unconditionally, and Hypothesis 3.1 (EQ(δ)) is equivalent to

$$\left| \mathbb{P}(P \text{ is a } B\text{-pair}) - \frac{1}{2} \right| \leq \frac{\delta}{4}(1 + o(1)),$$

for (L', P) uniform on X : the law of a single bit — whether rows 1, 2 lie in one column cycle of the marked pair $\{j, \sigma^\alpha(j)\}$ — of a single uniform square of $S(\lambda)$.

Proof. By (1) it suffices to prove $|J| = 2|X_A|$. The flip at an A -pair P whose columns lie in one row cycle at σ -distances (α, β) splits that cycle into an α - and a β -cycle and maps $S(\lambda) \rightarrow S(\mu)$; by Lemma 2.2 the pair P is still an A -pair afterwards, so $x \mapsto \text{unflip}(x)$ maps X_A into J_A . The flip at a fixed pair is an involution (the traded cycle again contains row 2, with the same row support), so $\text{join} = \text{flip}$ inverts it: $X_A \leftrightarrow J_A$ is a bijection. Composing with the switch of Lemma 2.3 gives a bijection $X_A \leftrightarrow J_B$, namely $x \mapsto \text{sw}(\text{unflip}(x))$ with inverse $(L, P) \mapsto \text{flip}(\text{sw}(L))$ — the map “switch-then-flip”, which is exactly the CGW joining step at a B -pair. Hence $|J| = |J_A| + |J_B| = 2|X_A|$. $\square$

3 The reduction theorem

Hypothesis 3.1 (B-pair equidistribution, EQ(δ)). *For every $\lambda \vdash n$ with a part $m = \alpha + \beta$, $\alpha \neq \beta$, $\alpha, \beta \geq 2$, and μ the split type, $|\bar{q}(\mu \rightarrow \lambda) - 1| \leq \delta(n)$, where $\bar{q}(\mu \rightarrow \lambda) := 2C_n(\lambda)/C_n(\mu) - 1 = \mathbb{P}_X[B]/\mathbb{P}_X[A]$ by Theorem 2.4.*

By Theorem 2.4, EQ(δ) is the statement $|\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{\delta}{4}(1 + o(1))$ for all adjacent pairs with $\alpha \neq \beta$. The proof below needs a tail bound on the number κ of cycles of σ ; we first prove it from the trivial half of Theorem 2.4 alone, so that nothing depends on the upper bound of [1, Lemma 3.12] (Remark 3.4).

Proposition 3.2 (the number of cycles, from the surviving direction only). *Let L be a uniform Latin square of order n , σ the column permutation of rows 1, 2, N_ℓ its number of ℓ -cycles and $\kappa = \sum_\ell N_\ell$. Then, using only the identity $|J| = 2|X_A|$ of Theorem 2.4 and $|X_A| \leq |X|$:*

(i) $\mathbb{E}[N_\ell] \leq 2n/(n - \ell)$ for every $\ell < n$; in particular $\mathbb{E}[N_\ell] \leq 4/\ell$ for $\ell \leq n/2$.

(ii) For every $k \geq 1$, with $(\kappa)_k = \kappa(\kappa - 1) \cdots (\kappa - k + 1)$,

$$\mathbb{E}[(\kappa)_k] \leq 2^{k-1} \sum_{\ell_1, \dots, \ell_k \geq 2} \frac{\mathbb{E}[mN_m]}{\ell_1 \cdots \ell_k} \Big|_{m=\ell_1+\dots+\ell_k} \leq 2^{k-1} \left(4 + \frac{k^2}{\log n}\right) (\log n)^k.$$

(iii) For $A \geq 3$, $\mathbb{P}[\kappa \geq A \log n] \leq (7 + \log n) n^{-\log((A-1)/2)}$, which is $O(n^{-2} \log n)$ for $A \geq 16$; and $\mathbb{P}[\text{all cycles of } \sigma \text{ have the same length } \ell < n] \leq 4d(n)/n$, $d(n)$ the number of divisors.

Proof. (i) For $\beta \neq \ell$ the joining instances $(L, \{j, j'\})$ with j in an ℓ -cycle and j' in a β -cycle number $\ell N_\ell \beta N_\beta$ per square, the splitting instances (pairs at distance ℓ on an $(\ell + \beta)$ -cycle) number $(\ell + \beta) N_{\ell+\beta}$, and $|J| = 2|X_A| \leq 2|X|$ gives $\mathbb{E}[\ell N_\ell \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta) N_{\ell+\beta}]$; for $\beta = \ell$ the same count with $\ell^2 \binom{N_\ell}{2}$ and $\ell N_{2\ell}$ gives $\mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] \leq 4\ell \mathbb{E}[N_{2\ell}]$. Summing the first over $\beta \neq \ell$ ($\sum_\beta \beta N_\beta = n$) and adding the second (its right side is the missing summand $m = 2\ell$): $\mathbb{E}[\ell N_\ell (n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] \leq 2\mathbb{E}[\sum_{m \geq \ell+2} m N_m] \leq 2n$. Hence $n\mathbb{E}[\ell N_\ell] = \mathbb{E}[\ell N_\ell (n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] + \ell \mathbb{E}[\ell N_\ell] \leq 2n + \ell \mathbb{E}[\ell N_\ell]$.

(ii) Count the instances $(L, (C_1, j_1), \dots, (C_k, j_k))$: C_i pairwise distinct σ -cycles of L of lengths ℓ_i and $j_i \in C_i$; there are $\ell_1 \cdots \ell_k$ of them per ordered k -tuple of distinct cycles with these lengths. Join C_1 and C_2 at the pair $\{j_1, j_2\}$ by a joining step: flip if the pair is an A -pair; if it is a B -pair, first trade the row cycle C_2 of j_2 (a switch in the sense of Lemma 2.3, whose proof applies to the row cycle through either column of the pair; it inverts C_2 in place and does not touch the cycle of j_1), then flip. The flip replaces σ by $\sigma \circ (j_1 j_2)$ on rows 1, 2, so the result has j_1, j_2 on one cycle of length $\ell_1 + \ell_2$ with $j_2 = \sigma^{\ell_2}(j_1)$ whether or not the trade was applied (CGW Lemma 3.3), every other cycle of σ (in particular $C_3, \dots, C_k$) untouched, and the map is at most two-to-one (the backflip at $\{j_1, j_2\}$ returns L or the traded square). Repeat with $\{j_1, j_3\}, \dots, \{j_1, j_k\}$, always trading the cycle of the new column when needed, so that the cycle containing j_1 is never inverted: after step i every earlier j_r is shifted by exactly ℓ_{i+1} . After $k - 1$ steps all j_i lie on one cycle of length $m = \sum \ell_i$ at the positions $j_i = \sigma^{\ell_i + \dots + \ell_k}(j_1)$, determined by j_1 and $(\ell_1, \dots, \ell_k)$, and the composite is at most 2^{k-1} -to-one. (With CGW's min-symbol rule the trade may invert the merged cycle and the positions are not determined; the count would then be 2^{2k-3} -to-one.) Hence the number of instances with lengths (ℓ_i) is at most $2^{k-1} \sum_{L'} m N_m(L')$, i.e. $\mathbb{E}[\ell_1 \cdots \ell_k T(\ell_1, \dots, \ell_k)] \leq 2^{k-1} \mathbb{E}[m N_m]$ with T the number of

ordered k -tuples of distinct cycles with these lengths; summing over (ℓ_i) gives the first bound, since $\sum_{(\ell_i)} T = (\kappa)_k$. For the second, split at $m \leq n/2$, where $\mathbb{E}[mN_m] \leq 4$ by (i) and $\sum_{m \leq n/2} \sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq (\sum_{\ell \leq n/2} \ell^{-1})^k \leq (\log n)^k$, and $m > n/2$, where some $\ell_i \geq m/k$, so $\sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq k \cdot (k/m)(\log n)^{k-1}$, and $\sum_{m > n/2} mN_m \cdot k^2(\log n)^{k-1}/m \leq k^2(\log n)^{k-1}$ because at most one cycle is longer than $n/2$.

(iii) Markov for $(\kappa)_k$ with $k = \lceil \log n \rceil \leq \log n + 1$: on $\{\kappa \geq K\}$, $(\kappa)_k \geq (K - k + 1)^k \geq ((A - 1) \log n)^k$ for $K = A \log n$, and $k^2 / \log n \leq \log n + 3$, so $\mathbb{P}[\kappa \geq K] \leq 2^{k-1}(4 + k^2 / \log n)(\log n)^k / ((A - 1) \log n)^k \leq (7 + \log n)(2/(A - 1))^k \leq (7 + \log n)(2/(A - 1))^{\log n}$ for $A \geq 3$, and $(A - 1)/2 \geq e^2$ for $A \geq 16$. For the equal-length types, $\ell \mid n$ and $\ell < n$ force $\ell \leq n/2$, so $\mathbb{P}[N_\ell = n/\ell] \leq \mathbb{E}[N_\ell] \ell / n \leq 4/n$; $\ell = n$ is the n -cycle. $\square$

Theorem 3.3. Assume $\text{EQ}(\delta)$ with $\delta = \delta(n) \rightarrow 0$. Then

$$d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)}).$$

In particular $\text{EQ}(o(1/\log n))$ implies Conjecture 1.1.

Proof. Call λ *plain* if not all its parts are equal (so the only non-plain types are (ℓ^κ) , $\kappa \geq 2$).

Routing. Every plain λ with $\kappa = \kappa(\lambda)$ parts is reachable from (n) by a chain of $\kappa - 1$ splits, each with $\alpha \neq \beta$. Build the chain in reverse, splitting parts of λ off a shrinking residual, in a suitable order; a collision means splitting a part p off a residual R with $p = R - p$. If the two largest parts of λ are unequal, split smallest-first and let them be the final pair: while at least three parts remain, the split-off minimum p satisfies $R - p \geq 2p > p$ (all parts are ≥ 2 and at least two other parts remain), so no collision occurs, and the final split is unequal by hypothesis. If instead the two largest parts are equal, say to $\ell_{\max}$, then $\ell_{\max} \neq n/2$ (else $\lambda = (n/2, n/2)$ is not plain), so we may split one copy of $\ell_{\max}$ off first without collision, then proceed smallest-first among the remaining parts, reserving for the final split the pair $(b, \ell_{\max})$ where $b < \ell_{\max}$ is the smallest part: at every intermediate step the residual minus the split-off minimum still contains the reserved $\ell_{\max}$ and the reserved b , hence exceeds the minimum (also when a further copy of $\ell_{\max}$ is the split-off minimum: $R - p \geq \ell_{\max} + b > p$), and the final split is unequal.

For each step of such a chain, Theorem 2.4 and $\text{EQ}(\delta)$ give $|S(p_{i+1})|/|S(p_i)| = (\gamma(p_{i+1})/\gamma(p_i))(1 + \varepsilon_i)$ with $|\varepsilon_i| \leq \delta$. Multiplying along the chain,

$$\frac{P_n(\lambda)}{P_n((n))} = \frac{\gamma(\lambda)}{\gamma((n))} \prod_{i=1}^{\kappa-1} (1 + \varepsilon_i)^{\pm 1}, \quad |\varepsilon_i| \leq \delta,$$

so for all plain λ with $\kappa(\lambda) \leq K$, $P_n(\lambda) = Q_n(\lambda) Z^{-1}(1 + O(K\delta))$ (the bound is vacuous unless $K\delta = O(1)$, which is the case of interest) where $Z > 0$ is a normalising constant independent of λ (namely $Z = \sum_{\nu} \gamma(\nu) C_n(\nu) / (|\mathcal{D}_n| C_n((n)))$).

Let $K = \lceil A \log n \rceil$ with A a large absolute constant, and let $\mathcal{B}$ be the set of types that are either non-plain or have $\kappa > K$. Under Q_n , Markov's inequality with the generating function $\sum_{\pi \in \mathcal{D}_n} 2^{\kappa(\pi)} = n! [x^n] e^{-2x} (1 - x)^{-2} \leq n \cdot n!$ (cf. [1, proof of Lemma 4.4]) gives $Q_n(\kappa > K) \leq e n 2^{-K} = O(n^{-2})$ for $A \log 2 \geq 3$; also $Q_n(\text{non-plain}) \leq \sum_{\ell \mid n, \ell < n} \gamma((\ell^{n/\ell})) / D_n = O(n^{-1})$. Under P_n , Proposition 3.2 gives $P_n(\kappa > K) = O(n^{-2} \log n)$ for $A \geq 16$ and $P_n(\text{non-plain}) = O(n^{-1+o(1)})$ without any input from [1] beyond the identity of Theorem 2.4.

On the complement of $\mathcal{B}$, summing $P_n = Q_n Z^{-1}(1 + O(K\delta))$ over $\lambda \notin \mathcal{B}$ and using the tail bounds shows $Z^{-1} = 1 + O(K\delta) + O(n^{-1+o(1)})$, whence $\sum_{\lambda \notin \mathcal{B}} |P_n - Q_n| = O(K\delta) + O(n^{-1+o(1)})$. Combining with the tails, $d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)})$. $\square$

Remark 3.4 (the gap in CGW’s Lemma 3.12). In [1], the direction $|S(\lambda, F)|N_\lambda \geq \frac{1}{2}|S(\mu, F)|N_\mu$ of Lemma 3.12 uses only that each splitting pair carries at most two edges and each joining pair at least one; by Theorem 2.4 it is the trivial inequality $|X| \geq |X_A|$. The direction $\leq \frac{3}{2}$ requires every splitting pair of every λ -square to carry exactly two edges that are also edges of the joining multigraph, and case 3 of the splitting procedure fails to supply them for every split other than $(2, 2)$ [2]. Consequently the upper bound of CGW’s Theorem 3.13, the lower bound of their Lemma 4.1, the parity bounds of Theorem 4.3, the bound $\mathbb{P}[\sigma \text{ is an } n\text{-cycle}] \leq 2n^{-2/3}$ of Lemma 4.4, both halves of Corollary 4.5, and the results of their Section 4.2 that rest on these (Corollary 4.6, Lemma 4.7, Theorems 4.9 and 4.11, Corollary 4.10 — the last superseded by [6]) are not established by the published argument. The *statements* of Theorem 4.9 (at most $9\sqrt{n}$ cycles with probability $1 - o(e^{-\sqrt{n}/4})$) and of the upper half of Corollary 4.5 (with $n/2$ in place of $n^{1/3}$) do follow from the surviving direction: Proposition 3.2(ii) with $k = \sqrt{n}$ gives $\mathbb{P}[\kappa \geq 9\sqrt{n}] \leq \exp(-(\frac{1}{2} - o(1))\sqrt{n} \log n)$, and $C_n(\lambda) \leq 2^{\kappa-1}C_n((n))$ gives $P_n(\lambda) \leq \frac{n}{2} \prod_i (2/c_i)$ for a type with parts c_i ; this repairs the use of both in [5, Thm. 9]. The surviving direction gives no upper bound on coarse types: in particular we know of no proof that two rows of a random Latin square form a single n -cycle with probability $o(1)$ (Lemma 4.4’s $2n^{-2/3}$; the second-moment method with [6] and Proposition 3.2(i) gives only $\leq \frac{5}{6} + o(1)$), and the parity theorem of [4] uses the unproved direction at odd splits (details in [2]). For $\min(\alpha, \beta) = 2$ the count can be corrected with the constant 2 in place of $\frac{3}{2}$ [2, Prop. 3]: for marks at distance 2 or $m - 2$ on an m -cycle, $C_n(\lambda)/C_n(\mu) \leq 2$, i.e. $\mathbb{P}_X[A] \geq \frac{1}{4}$ unconditionally. The present paper uses none of the affected statements of [1]: Proposition 3.2 replaces Corollary 4.5 in the proof of Theorem 3.3. The exact completion counts tabulated in [1] give $C_n(\lambda)/C_n(\mu) \in [0.985, 1.0035]$ for every adjacent pair at $7 \leq n \leq 11$ (and $= \frac{3}{2}$ at $n = 5$, $\frac{7}{8}, \frac{11}{14}, \frac{21}{22}$ at $n = 6$), so the statement itself is not in doubt.

4 The ladder identity

Notation: $(L, P) \in X = X(n, \lambda; \alpha, \beta)$, $P = \{j, j'\}$, frame $\pi = \pi_P$ ($L(\pi(r), j') = L(r, j)$), C_r the π -cycle of row r , $\rho_{x,y}$ the column permutation of rows x, y ($L(y, \rho_{x,y}(q)) = L(x, q)$), (x, y) *flippable* iff j, j' lie in different cycles of $\rho_{x,y}$, and the j' -trade at (x, y) exchanges rows x, y on the $\rho_{x,y}$ -cycle through j' , which replaces π by $(xy) \circ \pi$ if (x, y) is flippable and by $(xy) \circ \pi \circ (xy)$ otherwise. $B = \{1 \sim 2\}$ (rows 1, 2 in a common frame cycle), A its complement, $\mathbf{1}_B$ the bit; for B -instances d_{12} and d_{21} are the numbers of π -steps from 1 to 2 and from 2 to 1.

Definition 4.1 (ladder). Put $K_0 = \min(|C_1|, |C_2|)$ on A and $K_0 = \min(d_{12}, d_{21})$ on B , and for $1 \leq k < K_0$ let

$$x_k = \pi^{-k}(1), \quad y_k = \pi^{-k}(2), \quad T_k = \text{the } j'\text{-trade at } (x_k, y_k).$$

The pairs $\{x_k, y_k\}$ are the *ladder* of (L, P) , and $\text{Flip}(L, P) = \{k < K_0 : (x_k, y_k) \text{ flippable}\}$.

The rows $x_1, \dots, x_{K_0-1}$ are distinct rows of $C_1 \setminus \{1\}$ (on A) or of the arc from 2 to 1 (on B), the y_k likewise in $C_2 \setminus \{2\}$ or the arc from 1 to 2; so the $2(K_0 - 1)$ ladder rows are distinct and avoid rows 1, 2, and $K_0 \geq 2$: on A because π is a derangement, on B by the admissibility of the mark ($\pi(1) \neq 2, \pi(2) \neq 1$).

Theorem 4.2 (ladder identity). *For every n, λ and split $\alpha + \beta = m$ with $\alpha, \beta \geq 2$,*

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset], \quad \text{hence} \quad |\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{1}{2} \mathbb{P}_X[\text{Flip} = \emptyset].$$

The same identity holds with X replaced by the set of squares with a fixed pair of rows 1, 2 (of type λ) and a fixed mark P .

Proof. Fix $(L, P) \in X$ and write $K = \{1, \dots, K_0 - 1\}$. The ladder depends on (L, P) , so the T_k are not maps of X ; the statement to prove is that $(L, P) \sim T_S(L, P)$, $S \subseteq K$ (the trades at the unordered pairs of S , in any order), is an equivalence relation on X whose classes have exactly $2^{|K|}$ elements and on which the bit is $\mathbf{1}_B(T_S(L, P)) = \mathbf{1}_B(L, P) + |S \cap \text{Flip}| \bmod 2$. This follows once we know that each T_k is an involution of X which preserves K_0 , the set of unordered ladder pairs and the flippability of each of them, that T_k toggles the bit exactly when $k \in \text{Flip}$, and that trades at disjoint row pairs commute: then the same trades, at the same unordered pairs with the same flippabilities, are available from every point of a class (symmetry and transitivity), the $2^{|K|}$ points $T_S(L, P)$ are distinct (rows x_k, y_k are changed by T_k and by no other T_i), and the bit formula holds. If $\text{Flip} \neq \emptyset$ exactly half of the subsets S have $|S \cap \text{Flip}|$ odd, so B and A each contain half of the class; summing over classes gives the identity, and the bound follows from $\mathbb{P}[B] + \mathbb{P}[A] = 1$.

T_k is a well-defined involution of X . The j' -trade at (x_k, y_k) changes only rows $x_k, y_k \notin \{1, 2\}$, so σ and the mark are untouched and (L, P) stays in X ; it is an involution because the traded columns are again the ρ'_{x_k, y_k} -cycle through j' .

Effect on the frame. If $k \in \text{Flip}$ the new frame is $\pi' = (x_k y_k) \circ \pi$. On A this merges C_1 and C_2 : the new cycle runs $1 \rightarrow \pi(1) \rightarrow \dots \rightarrow x_{k+1} \rightarrow y_k \rightarrow \pi(y_k) \rightarrow \dots \rightarrow 2 \rightarrow \dots \rightarrow y_{k+1} \rightarrow x_k \rightarrow \dots \rightarrow 1$, so $d'_{12} = |C_1|$ and $d'_{21} = |C_2|$ (reading $x_{|C_1|} = 1, y_{|C_2|} = 2$). On B , x_k lies on the arc from 2 to 1 and y_k on the arc from 1 to 2 (as $k < \min(d_{12}, d_{21})$), and the split into $(x_k, \pi(x_k), \dots, \pi^{-1}(y_k))$ and $(y_k, \dots, \pi^{-1}(x_k))$ puts 1 (k steps after x_k , fewer than the d_{12} steps from x_k to y_k) in the first cycle and 2 in the second: the new state is in A with $|C'_1| = d_{12}$, $|C'_2| = d_{21}$. In both cases K_0 is preserved and the bit toggles. If $k \notin \text{Flip}$ the new frame is $(x_k y_k) \circ \pi \circ (x_k y_k)$: the cycle partition is unchanged up to exchanging the labels x_k, y_k , so K_0 and the bit are preserved.

The ladder is preserved. Let $\pi' = (x_k y_k) \circ \pi$ (the conjugation case is similar and easier). Since $\pi'^{-1}(r) = \pi^{-1}((x_k y_k)(r))$, the preimage chains of 1 and 2 under π' agree with those under π up to depth k and are exchanged beyond it: $\pi'^{-i}(1) = x_i$ and $\pi'^{-i}(2) = y_i$ for $i \leq k$, while $\pi'^{-(k+1)}(1) = \pi^{-1}(y_k) = y_{k+1}$ and then $\pi'^{-i}(1) = y_i$, $\pi'^{-i}(2) = x_i$ for $k < i < K_0$. So the unordered pairs $\{x_i, y_i\}$, $i \in K$, are the ladder of $T_k(L, P)$. In the conjugation case the same computation with $(x_k y_k) \circ \pi \circ (x_k y_k)$ gives $\pi'^{-i}(1) = (x_k y_k)(x_i)$, which again permutes the pairs $\{x_i, y_i\}$ among themselves (only $i = k$ is affected, by exchanging x_k, y_k).

Flippability is preserved and the T_k commute. T_k changes rows x_k, y_k only, and the flippability of (x_i, y_i) , $i \neq k$, depends only on rows x_i, y_i , which are different rows. Likewise T_i and T_k act on disjoint pairs of rows, each along a column set determined by its own two rows, so $T_i T_k = T_k T_i$. (After T_k the pair $\{x_i, y_i\}$ may have changed its index, but the trade at an unordered pair does not depend on the index or on the order of the two rows.)

The statement for fixed R holds because every T_k fixes rows 1 and 2. $\square$

Corollary 4.3 (the conjecture reduces to the ladder). *Write $f(n, \lambda; \alpha, \beta) = \mathbb{P}_X[\text{Flip} = \emptyset]$. By Theorem 2.4, $\bar{q}(\mu \rightarrow \lambda) - 1 = (\mathbb{P}_X[B] - \mathbb{P}_X[A])/\mathbb{P}_X[A]$ and $\mathbb{P}_X[A] \geq (1 - f)/2$, so $|\bar{q} - 1| \leq 2f/(1 - f)$: EQ(δ) holds with $\delta(n) = \max_{\lambda, \alpha \neq \beta} 2f/(1 - f)$. If this $\delta(n) \rightarrow 0$, Theorem 3.3 gives*

$$d_{\text{TV}}(P_n, Q_n) = O\left(\log n \cdot \max_{\lambda, \alpha \neq \beta} f(n, \lambda; \alpha, \beta) + n^{-1+o(1)}\right).$$

In particular the weak CGW conjecture follows from

$$(L) \quad \mathbb{P}_X[\text{no ladder pair is flippable}] = o(1/\log n) \quad \text{uniformly in } \lambda, \alpha, \beta. \quad (2)$$

Remark 4.4 (data). On Jacobson–Matthews samples with a uniformly random mark (9000, 7200, 3000 ladder-pair instances at $n = 30, 50, 100$; runs/s13/f1/): $\mathbb{P}[\text{Flip} = \emptyset \mid K_0] = 2^{-(K_0-1)}$ within statistical error for $K_0 \leq 6$ and consistent with it beyond; $\mathbb{P}[\text{Flip} = \emptyset] = 0.125, 0.079, 0.041 \approx 4/n$; the ladder pairs are flippable with frequency 0.5002 (62082 pairs at $n = 50$); and $\mathbb{P}[B] - \mathbb{P}[A] = -0.022 \pm 0.011, +0.008 \pm 0.012, -0.030 \pm 0.018$. On completions of a fixed rectangle (Remark 6.14; $7 \cdot 10^5$ marked instances at $n = 30$) $\mathbb{P}[\text{Flip} = \emptyset \mid K_0 - 1 \geq K] = 2^{-K}$ within noise to $K = 12\text{--}15$, in every tested type.

4.1 Offset ladders and short arcs

The ladder is one of a family. For an integer c (an *offset*) put

$$K_c = \begin{cases} \min(|C_1| - c, |C_2|) & \text{on } A, \\ \min(d_{21} - c, d_{12}) & \text{on } B, \end{cases} \quad D_c = \{(x_{c+i}, y_i) = (\pi^{-(c+i)}(1), \pi^{-i}(2)) : 1 \leq i < K_c\},$$

so D_0 is the ladder and for $c > 0$ the x -row is c steps deeper than the y -row (x -depths $c + 1, \dots, c + K_c - 1$ stay inside $C_1 \setminus \{1\}$, resp. the arc from 2 to 1; y -depths $1, \dots, K_c - 1$ inside $C_2 \setminus \{2\}$, resp. the arc from 1 to 2). We take $c \geq 0$; the families with the y -row deeper are obtained by exchanging the roles of rows 1 and 2 (and of C_1, C_2, d_{12}, d_{21}) throughout.

Theorem 4.5 (offset ladders). *For every offset $c \geq 0$, on the domain $\mathcal{D}_c = \{K_c \geq 2\}$ (where $\mathcal{D}_c \neq \emptyset$) the j' -trades at the pairs of D_c are commuting involutions that preserve K_c , the unordered pairs of D_c and their flippabilities, and toggle the bit iff the traded pair is flippable. Hence, with Flip_c the set of flippable pairs of D_c ,*

$$\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = \mathbb{P}[B, \mathcal{D}_c, \text{Flip}_c = \emptyset] - \mathbb{P}[A, \mathcal{D}_c, \text{Flip}_c = \emptyset].$$

Proof. Identical to Theorem 4.2; only the bookkeeping changes. Let $\pi' = (x_{c+i} y_i) \circ \pi$. The preimage chain of 1 under π' agrees with that of π up to depth $c+i$ and then continues along the chain of 2: $\pi'^{-(c+i+t)}(1) = y_{i+t}$; symmetrically $\pi'^{-(i+t)}(2) = x_{c+i+t}$ ($t \geq 1$). So the pair of D_c at y -depth $i' > i$ is $\{y_{i'}, x_{c+i'}\}$, unchanged as a set, and the pairs at depth $i' < i$ are untouched: D_c is preserved. (The same computation shows that trading a pair at depths (k, k') sends a deeper pair at depths (K, K') to depths $(K' + k - k', K - k + k')$, whose offset is $2(k - k') - (K - K')$; it is preserved iff $K - K' = k - k'$, so only families of constant offset, i.e. the D_c , are preserved.) On A the trade merges C_1, C_2 into a cycle with $d'_{12} = |C_1| - (c + i) + i = |C_1| - c$ and $d'_{21} = |C_2| + c$, so $K'_c = \min(d'_{21} - c, d'_{12}) = \min(|C_2|, |C_1| - c) = K_c$; on B the pair separates rows 1, 2 ($c + i < d_{21}$ puts x_{c+i} on the arc from 2 to 1, $i < d_{12}$ puts y_i on the arc from 1 to 2) and gives $|C'_1| = d_{12} + c$, $|C'_2| = d_{21} - c$, again $K'_c = \min(d_{12}, d_{21} - c) = K_c$. No pair of D_c is frame-adjacent: $y_i = \pi(x_{c+i})$ would need $d_{21} = c - 1$ (or $\pi(1) = 2$ when $c = 0$) and $x_{c+i} = \pi(y_i)$ would need $d_{21} = c + 1$, both excluded on $\mathcal{D}_c$. Commutation, preservation of flippability and the orbit count are as before. $\square$

Exact enumeration at $n = 7$ ($\lambda = (7)$, $\{\alpha, \beta\} = \{2, 5\}$) confirms the identities for $c = 0, 1, 2, 3$ to all digits ($\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = -0.005848, +0.001462, +0.003509, -0.001316$, equal to the right-hand sides).

Using the offsets to price short arcs. Fix $\ell \geq 2$ and consider B -instances with $d_{12} = \ell$. For $0 \leq c \leq n/2 - \ell$ and $d_{21} \geq c + \ell$ one has $K_c = \ell$, so D_c has $\ell - 1$ pairs, and the offset- c trade at a flippable pair of D_c lands in A with $|C_1| = \ell + c \leq n/2$ and $|C_2| = d_{21} - c \geq \ell$. Within a class of $\mathcal{D}_c$ all B -members have the same (d_{12}, d_{21}) and all A -members the same $(|C_1|, |C_2|) = (d_{12} + c, d_{21} - c)$, and Flip_c is a class invariant, so Theorem 4.5 class by class and Proposition 4.6 give

$$\mathbb{P}[B, d_{12} = \ell, d_{21} \geq c + \ell, \text{Flip}_c \neq \emptyset] = \mathbb{P}[A, |C_1| = \ell + c, |C_2| \geq \ell, \text{Flip}_c \neq \emptyset] \leq \frac{18}{n - \ell - c - 1} \leq \frac{36}{n - 2}. \quad (3)$$

Averaging over c : if $\theta(L, P)$ denotes the fraction of offsets $c \in [0, n/2 - \ell]$ whose diagonal D_c contains a flippable pair, then for every $\theta_0 > 0$

$$\mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta \geq \theta_0] \leq \frac{36}{(n - 2)\theta_0}, \quad (4)$$

since on $|C_1| \geq n/2 + \ell$ one has $d_{21} = |C_1| - \ell \geq n/2 \geq c + \ell$ for every such c , so $\mathbb{E}[\theta \mathbf{1}_E]$ is the average over c of the left side of (3), and Markov's inequality applies. Together with the splice-in bound $\mathbb{P}[B, d_{12} = \ell, |C_1| < n/2 + \ell] \leq 4/(n - 2\ell + 1) \leq 8/n$ for $\ell \leq n/4$ (Proposition 4.7 below), this prices every short arc except on the event $\{\theta < \theta_0\}$.

Proposition 4.6 (short frame cycles through a marked row are rare). *For (L, P) uniform on X and every $2 \leq \ell \leq n/2$,*

$$\mathbb{P}[|C_1| = \ell] \leq \frac{18}{n - \ell - 1}, \quad \text{hence} \quad \mathbb{P}[|C_1| \leq \ell] \leq \frac{36\ell}{n} \quad (\ell \leq n/2 - 1),$$

and the same bounds hold for C_2 .

Proof. Let $E_\ell = \{(L, P) \in X : |C_1| = \ell\}$. (The case $C_1 = \{1, 2\}$ cannot occur: it would mean $\sigma(j) = j'$ and $\sigma(j') = j$, but $j' = \sigma^\alpha(j)$ lies in an m -cycle with $m \geq 4$.)

Forward moves. From $(L, P) \in E_\ell$ choose $y \in C_1 \setminus \{1, 2\}$ and $x \notin C_1 \cup \{2\}$: at least $(\ell - 2)(n - \ell - 1)$ pairs $((\ell - 1)(n - \ell - 1)$ if $2 \notin C_1$; for $\ell = 2$ this is $n - 3$). If (x, y) is flippable, apply the j' -trade at (x, y) (trade rows x, y along the $\rho_{x,y}$ -cycle $Q_{x,y}$ through j'); otherwise turn C_1 (which makes (x, y) flippable, as $y \in C_1 \not\equiv x$) and then trade $Q_{x,y}$ in the turned square. In both cases the resulting frame is $(xy) \circ \pi^\pm$ with $\pi^\pm$ equal to π or to π inverted on C_1 , so the cycle of row 1 becomes $C_1 \cup C_x$, of length $\ell + |C_x| > \ell$; rows 1, 2 are untouched by the trade, and the result is in X (no turn, or $2 \in C_1$) or in J (turn with $2 \notin C_1$).

Backward moves. A square $(L', P) \in X \cup J$ is reached from $(L, P) \in E_\ell$ by the move at (x, y) only if x, y lie in the cycle C'_1 of row 1 in L' , and undoing the trade (the same trade, since $\rho'_{x,y} = \rho_{x,y}^{-1}$ on $Q_{x,y}$) splits C'_1 into the cycles $\{x, \pi'(x), \dots, \pi'^{-1}(y)\}$ and $\{y, \pi'(y), \dots, \pi'^{-1}(x)\}$ of $(xy) \circ \pi'$, of which the one containing row 1 must have length exactly ℓ : given L' , that arc is determined by its position relative to row 1 (ℓ choices), and the arc determines $\{x, y\}$. With the binary choice “turn back or not” this gives at most 2ℓ preimages per (L', P) .

Counting. The map $((L, P), \text{move}) \mapsto ((L', P), \text{inverse move})$ is injective, so $|E_\ell|(\ell - 2)(n - \ell - 1) \leq 2\ell(|X| + |J|) \leq 6\ell|X|$, using $|J| = 2|X_A| \leq 2|X|$ (Theorem 2.4). For $\ell \geq 3$ this is the claim; for $\ell = 2$ the forward count is $n - 3$ and the backward count 4, giving $12/(n - 3)$. The bound for C_2 is proved by the same argument with the roles of rows 1 and 2 exchanged: a turn of $C_2 \not\equiv 1$ replaces σ by $(\sigma(j)\sigma(j')) \circ \sigma$, which also splits the m -cycle into lengths α, β and lands in J . $\square$

Proposition 4.7 (the splice-in bound). *For $2 \leq \ell \leq n/4$, $\mathbb{P}_X[B, d_{12} = \ell, |C_1| < n/2 + \ell] \leq 4/(n - 2\ell + 1) \leq 8/n$, and the same with d_{21} in place of d_{12} .*

Proof. Let E be the event. *Forward moves.* From $(L, P) \in E$ choose $y = \pi^i(1)$ with $1 \leq i \leq \ell - 1$ (a row strictly inside the arc from 1 to 2; $\ell - 1$ choices) and a row $x \notin C_1$ ($n - |C_1| \geq (n - 2\ell + 1)/2$ choices), and let C_x be the frame cycle of x , which contains neither 1 nor 2. If (x, y) is not flippable, turn C_x : this trades columns j, j' on the rows of C_x only, so it preserves X , inverts π on C_x (Lemma 2.2), fixes row y , and replaces $\rho_{x,y}$ by $\rho_{x,y} \circ (j j')$, which makes (x, y) flippable. Then apply the j' -trade at (x, y) , which replaces the frame by $(x y) \circ \pi^\pm$ ($\pi^\pm = \pi$, or π inverted on C_x): the arc from 1 to 2 becomes $1 \rightarrow \dots \rightarrow \pi^{-1}(y) \rightarrow x \rightarrow \dots \rightarrow y \rightarrow \dots \rightarrow 2$ with C_x occupying positions $i, \dots, i + |C_x| - 1$, so d_{12} grows by $|C_x|$, rows 1, 2 are untouched, and the result is again in X_B with the same mark. *Backward moves.* Given the target $(L', P) \in X_B$ and ℓ , a preimage is determined by the position $i \in \{1, \dots, \ell - 1\}$ of x on the arc of L' from 1 to 2 (then $|C_x| = d'_{12} - \ell$ and $y = \pi^{i+|C_x|}(1)$ are determined), together with the binary choice of whether C_x was turned; undoing is the same j' -trade (the pair stays flippable, $\rho'_{x,y} = \rho_{x,y}^{-1}$ on the traded cycle) followed by the turn. So each target has at most $2(\ell - 1)$ preimages while each source has at least $(\ell - 1)(n - 2\ell + 1)/2$ forward moves, whence $|E|(\ell - 1)(n - 2\ell + 1)/2 \leq 2(\ell - 1)|X|$. $\square$

5 The witness hypothesis

Notation as in §4: $(L, P) \in X$, $P = \{j, j'\}$, frame π , ladder pairs (x_k, y_k) , K_0 , and for B -instances the offset diagonals D_c of §4.1. The statement below is the reference hypothesis of this paper — the least demanding sufficient condition for (L) that we have; the orbit method of §6 gives two others.

Say that a pair of rows (x, y) , $x, y \notin \{1, 2\}$, carries a *witness* if the $\rho_{x,y}$ -cycle through column j has length at most $\ell' = \log^5 n$ and does not contain j' (then (x, y) is flippable). A frame-adjacent pair ($y = \pi(x)$, so $\rho_{x,y}(j) = j'$) never carries one; the families below contain no such pair. For a pair of rows of a random Latin square the length of the j -cycle is spread like a uniform variable on $\{2, \dots, n\}$ (Remark 5.4 below), so a witness has probability $\approx \ell'/n$.

Hypothesis 5.1 (witness lemma). *Let (L, P) be uniform on X , uniformly in λ, α, β . Let $\mathcal{G}$ be one of the following families of blocks of pairs of rows, determined by the frame: (i) the ladder, each pair a block of its own; (ii) for a B -instance, with $\ell := d_{12}$, provided $\ell \leq n/4$ and $|C_1| \geq n/2 + \ell$, the diagonals D_c , $0 \leq c \leq n/2 - \ell$, each a block of $\ell - 1$ pairs (one family, not one per ℓ). Let $G_{\mathcal{G}}$ be the number of blocks containing a pair that carries a witness, and $\mu_{\mathcal{G}} = \sum_{B \in \mathcal{G}} \min(\frac{1}{2}, \ell' |B|/n)$. Then, for the ladder (i),*

$$\mathbb{P}[G_{\mathcal{G}} = 0 \text{ and } \mu_{\mathcal{G}} \geq \log^2 n] = o(1/\log n),$$

and for the diagonals (ii) there is a constant $\kappa_1 > 0$ such that

$$\mathbb{P}[G_{\mathcal{G}} < \kappa_1 \mu_{\mathcal{G}} \text{ and } \mu_{\mathcal{G}} \geq \log^2 n] = o(1/\log n).$$

(This is annealed: an exceptional set of frames of probability $o(1/\log n)$ is allowed. For a single pair of non-adjacent rows the witness probability is $\approx \ell'/n$, Remark 5.4, so $\kappa_1 \mu_{\mathcal{G}}$ is a constant fraction of the expected count if blocks are nearly independent given their shared rows.)

Proposition 5.2 ((L) follows from the witness lemma). *Hypothesis 5.1 implies (L), hence the weak CGW conjecture.*

Proof. Put $\ell_0 = n/\log^2 n$. By Theorem 4.2, $|\mathbb{P}[B] - \mathbb{P}[A]| \leq \mathbb{P}[\text{Flip} = \emptyset] \leq \mathbb{P}[K_0 \geq \ell_0, \text{Flip} = \emptyset] + \mathbb{P}[A, K_0 < \ell_0] + \mathbb{P}[B, K_0 < \ell_0]$.

Long ladders. On $\{K_0 \geq \ell_0\}$ take family (i): $\mu \geq (\ell_0 - 1)\ell'/n \geq \log^2 n$, so $G \geq 1$ except on an event of probability $o(1/\log n)$, and $G \geq 1$ means $\text{Flip} \neq \emptyset$.

Short cycles. $\mathbb{P}[A, K_0 < \ell_0] \leq 72\ell_0/n = 72/\log^2 n$ by Proposition 4.6.

Short arcs. $\mathbb{P}[B, K_0 < \ell_0] \leq \sum_{\ell < \ell_0} (\mathbb{P}[B, d_{12} = \ell] + \mathbb{P}[B, d_{21} = \ell])$; we treat $d_{12} = \ell$; the case $d_{21} = \ell$ reduces to it by exchanging rows 1 and 2 (with the symbol relabelling that restores the first row): this replaces σ by σ^{-1} , keeps the mark $\{j, j'\}$ with $j' = \sigma^{-\beta}(j)$, so maps $X(n, \lambda; \alpha, \beta)$ onto $X(n, \lambda; \beta, \alpha)$ measure-preservingly, conjugates the frame by (1 2), hence exchanges d_{12} with d_{21} , preserves the ladder as a set of unordered pairs and carries the diagonals with the x -row deeper to those with the y -row deeper, and leaves every $\rho_{x,y}$ with $x, y \notin \{1, 2\}$ unchanged; so the $d_{21} = \ell$ case for (λ, α, β) is the $d_{12} = \ell$ case of the hypothesis for (λ, β, α) , which “uniformly in λ, α, β ” covers (for $\alpha = \beta$ the hypothesis is read for the ordered mark). Fix ℓ . The part with $|C_1| < n/2 + \ell$ costs $\leq 8/n$ (Proposition 4.7). On $|C_1| \geq n/2 + \ell$ take family (ii), with $N = \lfloor n/2 - \ell \rfloor + 1 \geq n/4$ blocks: $\mu = N \min(\frac{1}{2}, (\ell - 1)\ell'/n) \geq \frac{1}{4} \min(n/2, (\ell - 1)\ell') \geq \log^2 n$, so except on an event F_ℓ of probability $o(1/\log n)$ the fraction of offsets whose diagonal contains a flippable pair is $\theta = G/N \geq \theta_0(\ell) := \kappa_1 \min(\frac{1}{2}, (\ell - 1)\ell'/n)$. By (4),

$$\begin{aligned} \sum_{\ell < \ell_0} \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta \geq \theta_0(\ell)] &\leq \sum_{\ell - 1 \leq n/(2\ell')} \frac{36n}{(n-2)\kappa_1(\ell-1)\ell'} + \sum_{n/(2\ell') < \ell - 1 < \ell_0} \frac{72}{(n-2)\kappa_1} \\ &= O\left(\frac{\log n}{\ell'} + \frac{\ell_0}{n}\right) = O\left(\frac{1}{\log^2 n}\right), \end{aligned}$$

and $\sum_{\ell < \ell_0} 8/n \leq 8/\log^2 n$. The exceptional events: the family (ii) is a function of the frame, and the events $\{d_{12} = \ell\}$ are disjoint, so $\sum_{\ell} \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, F_\ell] = \mathbb{P}[\bigcup_{\ell} \{B, d_{12} = \ell, |C_1| \geq n/2 + \ell, F_\ell\}]$ is the probability of a single failure event of the hypothesis, which is $o(1/\log n)$. Altogether $\mathbb{P}[\text{Flip} = \emptyset] = o(1/\log n)$. $\square$

Remark 5.3 (what the witness lemma is, and is not). (i) It is a statement about substructures of polylogarithmic size (a cycle of length $\leq \log^5 n$ of a row-pair permutation through a given column), in the space of completions of two rows and two columns, for families of pairs selected by the frame. It asks for a lower bound on their expected number and a lower-tail concentration (a second moment with relative variance $o(1/\log n)$ suffices). For two *fixed* rows, counts of such substructures are what the switching method of [6] controls; what it does not address is a family of row pairs selected by the square, and that is where the difficulty sits (§7). (ii) It asks nothing about long cycles, same-cycle events, or the fairness of any global statistic. (iii) Rigid squares ($\mathbb{Z}_p$ isotopes and their relatives, where no row pair has a short cycle) enter only through the failure probability, which must be $o(1/\log n)$; for them the hypothesis is simply the statement that they are rare. (iv) The constants: $\ell' = \log^5 n$ and $\ell_0 = n/\log^2 n$ are one admissible choice; the proof needs $\ell_0\ell'/n \geq \log^2 n$, $\ell' \geq 4\log^2 n$, $\ell_0/n = o(1/\log n)$ and $\log n/\ell' = o(1/\log n)$. (v) The block form is what the short-arc argument uses: the number of offsets with a witness, not the number of witnesses, since the diagonals of different offsets are dependent through the shared rows y_i . (vi) The hypothesis is stated only for the two families used; a version for arbitrary frame-determined families would be false, since frame-adjacent pairs never carry a witness.

Remark 5.4 (data). On Jacobson–Matthews samples with a uniformly random mark (`runs/s13/f1/`), the length of the j -cycle of ρ_{x_k, y_k} for a ladder pair is spread like a uniform variable on $\{2, \dots, n\}$:

$\mathbb{P}[\leq \ell] = 0.021, 0.081, 0.18, 0.38$ for $\ell = 2, 5, 10, 20$ at $n = 50$ against $(\ell - 1)/(n - 1) = 0.020, 0.082, 0.18, 0.39$, and $0.010, 0.041, 0.092, 0.19$ at $n = 100$; among instances with $K_0 \geq n/4$ a flippable ladder pair with j -cycle of length ≤ 10 exists in 93% ($n = 50$) and 95% ($n = 100$) of the instances, and one of length ≤ 20 in 99.6% and 99.8%. These samples see only typical types λ ; the hypothesis is required uniformly in λ . The fixed-rectangle sampler of Remark 6.14 reaches the rare types: there the witness events along the ladder are independent Bernoulli variables to three digits (flat hazard, $\mathbb{P}[\text{no witness in } K \text{ pairs}] = (1 - w_1)^K$ to $K = 40$ at $n = 100$) and the witness counts across the offset diagonals have binomial lower tails, identically for (n) and for $(4, 2^{(n-4)/2})$. Classes with all cycles of $\rho_{1,2}$ of length ≤ 3 carry no admissible mark and do not enter.

6 The orbit method: alternative sufficient conditions

Throughout this section $X = X(n, \lambda; \alpha, \beta)$, and “legal” means: preserves the type of rows 1, 2 and the mark, i.e. maps X to itself. The mark is now written $\{p, p'\}$ (so $p = j$, $p' = j'$ in the notation of §§2–4; the letters j, j' are freed for other columns), π is its frame, and a column pair (q, c) is *free* if $\{q, c\} \cap \{p, p'\} = \emptyset$. A pair of rows (x, y) is called *parallel* if it is flippable (p, p' in different cycles of $\rho_{x,y}$) and *crossed* otherwise; Q_p and $Q_{p'}$ denote the $\rho_{x,y}$ -cycles through p and p' . A free pair (q, c) is *separated* by $\{p, p'\}$ in $\rho_{x,y}$ if q and c lie on different arcs of a common cycle through p, p' (crossed case) or one in Q_p and the other in $Q_{p'} \neq Q_p$ (parallel case). The method bounds $\mathbb{P}_X[\text{Flip}_I = \emptyset]$ by a product of conditionally independent factors; the two hypotheses it leads to are compared with Hypothesis 5.1 after (c'), and in (e) the same mechanism, applied to the first ladder pair alone, gives an exact expression for $\mathbb{P}_X[B] - \mathbb{P}_X[A]$ and a third alternative hypothesis.

(a) Transposition: which half of the conditioning is the obstacle. Let L be uniform on *all* Latin squares of order n (no condition on rows 1, 2 in this paragraph), let x, y carry four distinct symbols on $\{p, p'\}$, and condition on the two full columns p, p' (the frame π of (p, p')). In the transposed square $L^\top$ the columns p, p' are the distinguished rows, the frame π is their column permutation, and the event “ p, p' lie in different cycles of $\rho_{x,y}$ ” is the event “ $\{x, y\}$ is an A -pair for the distinguished rows”. Conditioning on the two full rows p, p' of $L^\top$ is conditioning on their type and on the position of the pair $\{x, y\}$ in their cycle structure, which is free by symbol and column relabelling. Hence, by Lemma 2.3 ($|J_A| = |J_B|$, which needs no condition on the cycle lengths) and CGW’s Theorem 3.13 (the inequality $\mathbb{P}_X[A] \geq \frac{1}{3}$, whose published proof has a gap for every split other than $(2, 2)$, Remark 3.4):

$$\mathbb{P}[p, p' \text{ in different cycles of } \rho_{x,y} \mid \text{columns } p, p'] \begin{cases} = \frac{1}{2} & \text{if } x, y \text{ are apart in the frame,} \\ \geq \frac{1}{3} & \text{if together at distances } \geq 2. \end{cases} \quad (5)$$

(The second bound is CGW’s $\frac{3}{2}$, affected by the gap; it holds with $\frac{1}{4}$ unconditionally at distance 2, Remark 3.4.) So the statement with the *columns* prescribed is CGW’s theorem; it is the conditioning on rows 1, 2 — on their type and the mark, which is what defines X and cannot be removed — that is the obstacle. In the same transposed picture the ladder trades T_k of Theorem 4.2 at flippable pairs are CGW’s flips and backflips at the column pairs $\{x_k, y_k\}$ of $L^\top$, and Theorem 4.2 is exactly the legal half of CGW’s Lemma 3.12 for the distinguished rows p, p' : the other half — the switch and the cross-switch at $\{x_k, y_k\}$ — trades the columns p, p'

of L along a frame cycle or frame arc through exactly one of rows 1, 2 (apart: $C_1 \ni 1$, $C_1 \not\ni 2$; together: the arc from x_k to y_k contains 1 but not 2, the other arc 2 but not 1), so it changes the type of rows 1, 2 and leaves X . The switch is, in the original orientation, the unflip of rows 1, 2 at (p, p') (which maps $X_A \rightarrow J_A$); it toggles the crossed/parallel status of *every* pair (x, y) with x on the side of row 1 and y on the side of row 2, and reverses the orientation of one side, so it carries the ladder (a diagonal of the side-by-side array of pairs) to a different family (an anti-diagonal). The cross-switch likewise maps X_B to J_B (it splits the σ -cycle through p, p' while keeping rows 1, 2 in one frame cycle, CGW Lemma 3.10) and toggles the interior pairs. Both are exact but not usable for the ladder bits in X .

(b) Row trades are free in X ; column turns are conditionally free. For rows $z, w \notin \{1, 2\}$ every trade of rows z, w along a $\rho_{z, w}$ -cycle or $\rho_{z, w}$ -arc changes rows z, w only, hence is legal in X . In particular:

Lemma 6.1 (exact fairness of column-pair status for generic rows). *Let $x, y \notin \{1, 2\}$ and let (q, c) be a column pair lying in two different cycles of $\rho_{x, y}$. The CGW switch at (q, c) for the rows x, y (trade of rows x, y along the one of the two cycles that contains the smaller minimum symbol) is a legal involution of X that toggles whether x, y lie in one column cycle of (q, c) , and preserves the cycle partition of $\rho_{x, y}$ and the separation status of every column pair with respect to $\{p, p'\}$. Hence, conditionally on everything determined by the other rows and by the cycle partition of $\rho_{x, y}$,*

$$\mathbb{P}_X[x, y \text{ in different column cycles of } (q, c)] = \frac{1}{2}.$$

Proof. As in Lemma 2.3 with $(1, 2)$ replaced by (x, y) : the trade swaps the entries of rows x, y on the columns of the chosen cycle $Q \ni q$, $Q \not\ni c$, so $\delta_{q, c} \mapsto \delta_{q, c} \circ (xy)$, which splits or joins the column cycles through x, y . It inverts Q in place, so the symbol sets of the two cycles are unchanged and the min-symbol rule re-selects Q : an involution. Rows 1, 2 are untouched, so type and mark are preserved. Inverting Q exchanges the two arcs between p and p' when $Q \ni p, p'$, so separation of any pair by $\{p, p'\}$ is preserved. (If Q contains p or p' the frame changes at rows x, y ; so frame-determined data are part of the conditioning only when Q avoids p, p' .) $\square$

A column turn, by contrast, changes the rows of a column cycle. Let (q, c) be a free column pair and Z one of its column cycles.

Lemma 6.2 (legal column turns). *The turn of Z (swapping the entries of columns q, c in the rows of Z) is a measure-preserving involution of X if and only if $|Z \cap \{1, 2\}| \in \{0, 2\}$; when $|Z \cap \{1, 2\}| = 2$ it conjugates σ by (qc) and keeps the type and the mark. It preserves the columns p, p' (hence the frame and every frame-determined family of row pairs) and the column-cycle partition of (q, c) and of every column pair disjoint from $\{q, c\}$ (in particular of every pair of a matching containing (q, c)). For a row pair (x, y) with $|Z \cap \{x, y\}| = 1$ it replaces $\rho_{x, y}$ by $\rho_{x, y} \circ (qc)$ (if $x \in Z$) or $(qc) \circ \rho_{x, y}$ (if $y \in Z$), which toggles the crossed/parallel status of (x, y) exactly when (q, c) is separated by $\{p, p'\}$ in $\rho_{x, y}$ (different arcs when crossed, different cycles $Q_p, Q_{p'}$ when parallel), and leaves (q, c) separated afterwards; with $|Z \cap \{x, y\}| \in \{0, 2\}$ the status is unchanged. Turns of cycles of column pairs from a fixed matching $\mathcal{M}$ of the free columns pairwise commute.*

Proof. A column cycle is a Latin trade (Lemma 2.2), so the turn is a bijection of Latin squares and an involution, and it preserves the cycle partition of (q, c) ; column pairs disjoint from $\{q, c\}$

are untouched, and so are columns p, p' . Rows 1, 2 change iff Z meets them; if $Z \ni 1, 2$ both rows swap their entries at q, c , so $\sigma' = (qc)\sigma(qc)$, whose type is that of σ and in which p, p' (fixed by (qc)) keep their distance. If Z meets exactly one of rows 1, 2, $\sigma' = \sigma \circ (qc)$ or $(qc) \circ \sigma$, which changes the type. The effect on $\rho_{x,y}$: if $x \in Z$ and $y \notin Z$ the entries of row x at q, c are exchanged, so $\rho'_{x,y} = \rho_{x,y} \circ (qc)$, and symmetrically; multiplying by a transposition of two points on one cycle splits it at those points, so p, p' are separated iff q, c lie on different arcs between them, and multiplying by a transposition of two points on different cycles merges the cycles, so p, p' are joined iff q, c lie one in each of their cycles; in both cases q, c lie one on each side afterwards. Turns of cycles of disjoint column pairs act on disjoint sets of cells; two cycles of the same pair have disjoint row sets. $\square$

(c) The product bound. Fix a matching $\mathcal{M}$ of the free columns, independent of the square, and a frame-determined family $\mathcal{G}$ of row pairs (x, y) , $x, y \notin \{1, 2\}$, whose x -rows are pairwise distinct and disjoint from its y -rows (a block of the ladder, where the pairs are even disjoint; or a union of diagonals D_c with distinct x -rows, which may share y -rows). For $(x, y) \in \mathcal{G}$ call a column pair $(q, c) \in \mathcal{M}$ a *clean coordinate* of (x, y) if the column cycle Z of (q, c) through x has length at most ℓ_1 (a fixed constant), satisfies $|Z \cap \{1, 2\}| \in \{0, 2\}$, and contains no row of $\mathcal{G}$ other than x (no other x -row and no y -row, in particular not y). Let $\mathcal{T}_{x,y}$ be the set of clean coordinates of (x, y) and $\eta_{x,y} \in \{0, 1\}^{\mathcal{T}_{x,y}}$. By Lemma 6.2 the turns of all clean coordinates of all pairs of $\mathcal{G}$ commute, generate a group acting freely on X , preserve $\mathcal{G}$ and the sets $\mathcal{T}_{x,y}$, and

$$\rho_{x,y}(\eta) = \rho_{x,y}(0) \circ \prod_{(q,c) \in \mathcal{T}_{x,y}: \eta_{x,y}(q,c)=1} (qc),$$

the bit $b_{x,y}(\eta) = [(x, y) \text{ parallel}]$ depending on $\eta_{x,y}$ only. Let the *orbit weight* of (x, y) be

$$\bar{y}_{x,y} = \mathbb{E}_{\eta_{x,y}} \left[\frac{\#\{(q, c) \in \mathcal{T}_{x,y} : (q, c) \text{ separated by } \{p, p'\} \text{ in } \rho_{x,y}(\eta)\}}{|\mathcal{T}_{x,y}|} \right]$$

(0 if $\mathcal{T}_{x,y} = \emptyset$), a function of the orbit, and $Y_{\mathcal{G}} = \sum_{(x,y) \in \mathcal{G}} \bar{y}_{x,y}$.

Proposition 6.3 (product bound). *Conditionally on the orbit, the bits $b_{x,y}$, $(x, y) \in \mathcal{G}$, are independent, and*

$$\mathbb{P}[(x, y) \text{ crossed} \mid \text{orbit}] \leq 1 - \frac{1}{2} \bar{y}_{x,y}, \quad \mathbb{P}[\text{no pair of } \mathcal{G} \text{ is flippable} \mid \text{orbit}] \leq \prod_{(x,y) \in \mathcal{G}} (1 - \frac{1}{2} \bar{y}_{x,y}) \leq e^{-Y_{\mathcal{G}}/2}.$$

Consequently $\mathbb{P}_X[\text{Flip}_{\mathcal{G}} = \emptyset] \leq \mathbb{E}_X[e^{-Y_{\mathcal{G}}/2}]$.

Proof. The action is free: distinct column pairs act on disjoint cells, and two clean coordinates of different pairs at the same (q, c) are the cycles through two distinct x -rows, each avoiding the other, hence different cycles; the sets $\mathcal{T}_{x,y}$ are orbit-invariant by Lemma 6.2 (row sets of the cycles, their meeting $\{1, 2\}$, and the frame-determined family are preserved). The uniform measure on X restricted to an orbit is therefore uniform on the orbit, which is the product of the cubes $\{0, 1\}^{\mathcal{T}_{x,y}}$. A clean coordinate of (x, y) contains x and not y , and a clean coordinate of another pair contains neither x nor y (it avoids all x -rows but its own and all y -rows), so $b_{x,y}$ depends on $\eta_{x,y}$ alone, whence the independence. Fix (x, y) and $Z = (q, c) \in \mathcal{T}_{x,y}$. Whether Z is separated in $\rho_{x,y}(\eta)$ does not depend on η_Z (Lemma 6.2: the turn of Z leaves Z separated), and when it is, the two values of η_Z give opposite bits. Hence $\mathbb{P}[b_{x,y} = 0] \leq 1 - \frac{1}{2} \mathbb{P}_{\eta}[Z \text{ separated}]$ for every Z , and averaging the right side over $Z \in \mathcal{T}_{x,y}$ gives the first bound; the second is the product of the first over the independent bits. $\square$

Hypothesis 6.4 (orbit genericity; fixed matching). *Fix the matching $\mathcal{M}$ of the free columns in consecutive pairs (in the natural order of the columns) and a constant ℓ_1 as in (c), and for a family $\mathcal{G}$ and $D \subseteq \mathcal{G}$ write $Y_D = \sum_{(x,y) \in D} \bar{y}_{x,y}$ (orbit weights computed with cleanliness with respect to all rows of $\mathcal{G}$). There is a constant $\kappa_0 > 0$ such that, with $\ell_0 = n/\log^2 n$ and uniformly in λ, α, β : (i) for the sub-ladder I of the first $\min(K_0 - 1, \lceil n/8 \rceil)$ ladder pairs, $\mathbb{E}_X[e^{-Y_I/2}; K_0 \geq \ell_0] = o(1/\log n)$; (ii) for a B -instance, with $\ell := d_{12} < \ell_0$ and $|C_1| \geq n/2 + \ell$, $S = \{2\ell t : 0 \leq t \leq (n/2 - \ell)/(2\ell)\}$ and $\mathcal{G} = \bigcup_{c \in S} D_c$: the event that fewer than $\kappa_0|S|$ of the diagonals D_c , $c \in S$, have $Y_{D_c} \geq \kappa_0(\ell - 1)$ has probability $o(1/\log n)$ (one event, not one per ℓ); and symmetrically for d_{21} . (A fixed small fraction, not a half: with a fixed matching the orbit weight of a single pair vanishes with probability ≈ 0.7 , §(d), so for $\ell = 2$ only a minority of the one-pair diagonals can qualify.)*

Since $\bar{y}_{x,y}$ is typically of constant size (§(d): 0.16–0.19 on average with the fixed matching), (i) asks that the sum of $\approx n/8$ such weights not collapse; its content is a *genericity constant* — that a clean coordinate is separated by the mark with probability bounded below — which none of the exact tools of this paper provides (§7).

(c') Adaptive coordinates: the toggling is automatic. Proposition 6.3 needs the clean coordinates of a ladder pair to be *separated* by $\{p, p'\}$ with positive probability, a genericity input we cannot prove (§7). It disappears if the coordinates are chosen along $\rho_{x,y}$ from the mark. For a pair (x, y) of the family and $t \geq 1$ let

$$(q^t, c^t) := (\rho_{x,y}^t(p), \rho_{x,y}^t(p')),$$

and let $\mu_{x,y} := \min(d(p \rightarrow p'), d(p' \rightarrow p))$ if (x, y) is crossed and $\min(|Q_p|, |Q_{p'}|)$ if parallel. For $t < \mu_{x,y}$ the pair (q^t, c^t) is free ($q^t \neq c^t$, both $\notin \{p, p'\}$) and separated by $\{p, p'\}$ in $\rho_{x,y}$ by construction: on different arcs when crossed, one in Q_p and one in $Q_{p'}$ when parallel. So the turn of its column cycle Z through x (with $y \notin Z$) toggles the status of (x, y) whenever it is legal (Lemma 6.2). What makes these coordinates usable in the orbit method is that they are invariant under their own turn. (We keep (c) for comparison: its orbit weight is what the data of (d) measure, and it is where the genericity constant of §7 lives.)

Lemma 6.5 (the adaptive coordinate is orbit-invariant). *Let $t < \mu_{x,y}$, let Z be the column cycle of (q^t, c^t) through x with $y \notin Z$, and let L' be the turn of Z . Then $\rho'_{x,y} = \rho_{x,y} \circ (q^t c^t)$, the status of (x, y) toggles, $\mu'_{x,y} = \mu_{x,y}$, and $(\rho'^s(p), \rho'^s(p')) = (q^s, c^s)$ for every $s \leq t$; in particular the t -th candidate of L' is (q^t, c^t) again, and its column cycle through x is Z (same row set).*

Proof. Write $\rho = \rho_{x,y}$, $q = q^t$, $c = c^t$. The turn exchanges the entries of x at q, c (and those of the other rows of Z , which do not affect ρ), so $\rho' = \rho \circ (qc)$: $\rho'(q) = \rho(c)$, $\rho'(c) = \rho(q)$, $\rho' = \rho$ elsewhere. Crossed case, with $p = u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_a = p' \rightarrow v_1 \rightarrow \dots \rightarrow v_b = p$ and $q = u_t$, $c = v_t$ ($t < \min(a, b)$): the cycles of ρ' through p and p' are $u_0 \rightarrow \dots \rightarrow u_t \rightarrow v_{t+1} \rightarrow \dots \rightarrow v_b = p$ and $u_{t+1} \rightarrow \dots \rightarrow u_a = p' \rightarrow v_1 \rightarrow \dots \rightarrow v_t \rightarrow u_{t+1}$, of lengths b and a , so (x, y) is now parallel with $\mu' = \min(a, b) = \mu$, and the paths $p \rightarrow u_s$, $p' \rightarrow v_s$ for $s \leq t$ are untouched. Parallel case, $Q_p = (p = u_0, u_1, \dots, u_{a-1})$, $Q_{p'} = (p' = v_0, \dots, v_{b-1})$, $q = u_t$, $c = v_t$ ($t < \min(a, b)$): the single cycle of ρ' is $u_0 \rightarrow \dots \rightarrow u_t \rightarrow v_{t+1} \rightarrow \dots \rightarrow v_{b-1} \rightarrow v_0 = p' \rightarrow v_1 \rightarrow \dots \rightarrow v_t \rightarrow u_{t+1} \rightarrow \dots \rightarrow u_{a-1} \rightarrow u_0$, with $d'(p \rightarrow p') = b$, $d'(p' \rightarrow p) = a$, so crossed with $\mu' = \mu$, and again the paths up to step t are untouched. The row set of the turned cycle is preserved (Lemma 2.2). $\square$

Proposition 6.6 (orbit bound with adaptive coordinates). *Let $(L, P) \in X$, I a frame-determined family of pairs (x_k, y_k) as in §(c) (distinct x -rows, disjoint from the y -rows, none in $\{1, 2\}$; the ladder, or a union of offset diagonals), and $T, \ell_1 \geq 1$. Call the candidate (k, t) good if $t \leq \min(T, \mu_k - 1)$ and the column cycle Z_k^t of (q_k^t, c_k^t) through x_k has length $\leq \ell_1$, meets $\{1, 2\}$ in 0 or 2 rows and meets the rows of I only in x_k . Let $t_0(k)$ be the least good t (if any), $C_k = \{q_k^t, c_k^t : t \leq t_0(k)\}$ (all candidates $t \leq \min(T, \mu_k - 1)$ if there is none), and*

$$U = \{k \in I : t_0(k) \text{ exists and } \{q_k^{t_0(k)}, c_k^{t_0(k)}\} \cap C_l = \emptyset \text{ for all } l \in I \setminus \{k\}\}.$$

Then the turns T_k of $Z_k^{t_0(k)}$, $k \in U$, are legal, commute, generate a free action of $\{0, 1\}^U$ on X that preserves the frame, the family, every μ_l , $t_0(l)$, C_l ($l \in I$) and U ; T_k toggles the status of (x_k, y_k) and fixes that of every other pair. Hence

$$\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X[2^{-|U|}].$$

Proof. T_k changes only columns $q_k, c_k := q_k^{t_0(k)}, c_k^{t_0(k)}$ in the rows of Z_k (lengths of cycles are preserved with their row sets, so the length condition is invariant along with the rest). Legality: Z_k meets $\{1, 2\}$ in 0 or 2 rows and $\{q_k, c_k\} \cap \{p, p'\} = \emptyset$, so the type of rows 1, 2 and the mark are preserved (Lemma 6.2); columns p, p' are untouched, so the frame and the ladder are preserved. For $k \neq l$ in U the column pairs $\{q_k, c_k\}, \{q_l, c_l\} \subseteq C_l$ are disjoint, so the turns act on disjoint cells and commute, and the action is free. Invariance of the data of a pair $l \neq k$ under T_k : rows $x_l, y_l \notin Z_k$, so ρ_l , its status, μ_l and all its candidates are unchanged; the goodness of candidate (l, t) depends on the column cycle of (q_l^t, c_l^t) through x_l , i.e. on the contents of columns q_l^t, c_l^t , which are not in $\{q_k, c_k\}$ for $t \leq t_0(l)$ (resp. for all candidates if $t_0(l)$ does not exist) by the defining condition of U ; hence $t_0(l)$ and C_l are unchanged. Invariance of the data of k itself: by Lemma 6.5, μ_k and the candidates $t \leq t_0(k)$ are unchanged; the candidates $t < t_0(k)$ have columns different from q_k, c_k (distinct points of the two paths), so their cycles and non-goodness are unchanged, and Z_k itself keeps its row set, so $t_0(k)$ is still good; C_k is unchanged. If $Z_k \ni 1, 2$, rows 1, 2 change at q_k, c_k , which affects none of the above. Hence U is invariant, every orbit of the cube is a cube of size $2^{|U|}$ on which the status of pair $k \in U$ is $\text{bit}_k(L) \oplus \varepsilon_k$, and $\text{Flip}_I = \emptyset$ forces one value of $(\varepsilon_k)_{k \in U}$; X is uniform. $\square$

Hypothesis 6.7 (short clean cycles at the ladder rows). *There are $T, \ell_1 = O(\log^2 n)$ and a constant $\kappa_2 > 0$ such that, uniformly in λ, α, β , with $\ell_0 = n/(16 \log^2 n)$, $M = \lfloor n/\log^4 n \rfloor$ and U as in Proposition 6.6:*

- (i) *for the sub-ladder I of the first $\min(K_0 - 1, \lceil \ell_0 \rceil)$ ladder pairs, $\mathbb{E}_X[2^{-|U|}; K_0 \geq \ell_0] = o(1/\log n)$;*
- (ii) *for a B -instance, with $\ell := d_{12} < \ell_0$ and $|C_1| \geq n/2 + \ell$: let $S = \{2\ell t : 0 \leq t \leq (n/2 - \ell)/(2\ell)\}$ (offsets at gaps 2ℓ , so that the x -rows of the diagonals D_c , $c \in S$, are pairwise distinct; $|S| \geq n/(8\ell) \geq 2 \log^2 n$), let S' be the first $m_\ell := \min(|S|, \lceil n/(\log^2 n \cdot \min(\ell - 1, M)) \rceil)$ offsets of S , let $D_c^{(M)}$ be the first $\min(\ell - 1, M)$ pairs of D_c , and take U for the family $\mathcal{F}_\ell = \bigcup_{c \in S'} D_c^{(M)}$ (which has at most $2n/\log^2 n$ pairs). Then the event that fewer than $\kappa_2 \log^2 n$ of the diagonals $D_c^{(M)}$, $c \in S'$, meet U has probability $o(1/\log n)$ (one event, not one per ℓ); and symmetrically for d_{21} .*

The scaling in (ii) is forced: a diagonal of $\ell - 1$ pairs meets U with probability $\approx \min(\ell - 1, M) T \ell_1 / n$, so for short arcs only a small fraction of the diagonals can be expected to meet U , and the family must be kept to $O(n / \log^2 n)$ pairs for the cleanliness and collision conditions of Proposition 6.6 to have constant cost; $|S'| \min(\ell - 1, M) \approx n / \log^2 n$ makes the expected number of diagonals meeting U of order $\log^2 n$ for every ℓ .

Proposition 6.8. *Each of Hypotheses 6.4 and 6.7 implies (L), hence the weak CGW conjecture.*

Proof. By Theorem 4.2, $|\mathbb{P}[B] - \mathbb{P}[A]| \leq \mathbb{P}[\text{Flip} = \emptyset] \leq \mathbb{P}[K_0 \geq \ell_0, \text{Flip} = \emptyset] + \mathbb{P}[A, K_0 < \ell_0] + \mathbb{P}[B, K_0 < \ell_0]$, with ℓ_0 as in the hypothesis used. *Long ladders.* K_0 is a function of the frame, hence orbit-invariant, and by Proposition 6.3 resp. Proposition 6.6 $\mathbb{P}[\text{Flip}_I = \emptyset \mid \text{orbit}] \leq e^{-Y_I/2}$ resp. $2^{-|U|}$, so $\mathbb{P}[K_0 \geq \ell_0, \text{Flip} = \emptyset] = o(1/\log n)$ by (i). *Short ladders on A.* $\mathbb{P}[A, K_0 < \ell_0] \leq 72\ell_0/n$ by Proposition 4.6. *Short arcs on B.* Fix $\ell < \ell_0$ and consider B -instances with $d_{12} = \ell$; the case $d_{21} = \ell$ reduces to it by the involution exchanging rows 1, 2, as in Proposition 5.2. The part $|C_1| < n/2 + \ell$ costs at most $8/n$ by Proposition 4.7. On $|C_1| \geq n/2 + \ell$ the family of (ii) is as required by the propositions: its x -rows lie in the disjoint depth intervals $[c + 1, c + \ell - 1]$, $c \in S$, on the frame arc from 2 to 1, and its y -rows $\pi^{-i}(2)$ on the arc from 1 to 2. Given the orbit the statuses of all pairs are independent, and the diagonals' flippabilities are independent (they are disjoint sets of pairs). Under Hypothesis 6.4, a diagonal with $Y_{D_c} \geq \kappa_0(\ell - 1)$ is flippable with conditional probability $\geq 1 - e^{-\kappa_0(\ell-1)/2} \geq 4\theta_0$, $\theta_0 := \frac{1}{4}(1 - e^{-\kappa_0/2})$; off the exceptional event of (ii) at least $\kappa_0|S|$ diagonals qualify, so the number of flippable offsets has conditional mean $\geq 4\theta_0\kappa_0|S|$ and the fraction θ_S is $\geq 2\theta_0\kappa_0 =: \theta'_0$ except with conditional probability $\leq e^{-\theta_0\kappa_0|S|/2} \leq e^{-\theta_0\kappa_0 \log^2 n/16}$ (as $|S| \geq n/(8\ell_0) = \log^2 n/8$). Under Hypothesis 6.7, a diagonal $D_c^{(M)}$ meeting U is flippable with conditional probability $\geq \frac{1}{2}$ (the status of a pair $k \in U$ is $\text{bit}_k \oplus \varepsilon_k$ with ε uniform on the orbit); off the exceptional event of (ii) at least $\kappa_2 \log^2 n$ diagonals of S' meet U , so the number of flippable offsets in S' has conditional mean $\mu \geq \frac{1}{2}\kappa_2 \log^2 n$ and is $\geq \mu/2$ except with conditional probability $\leq e^{-\mu/8} \leq e^{-\kappa_2 \log^2 n/16}$; hence the fraction $\theta_{S'}$ is $\geq \theta_0(\ell) := \kappa_2 \log^2 n / (4m_\ell)$. In both cases the conditional failure probabilities are summable over $\ell < \ell_0$ to $o(1/n)$. The derivation of (4) from (3) applies to the average over any set of offsets determined by $\ell = d_{12}$ alone, such as S and S' , so

$$\sum_{\ell < \ell_0} \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta_S \geq \theta_0(\ell)] \leq \sum_{\ell < \ell_0} \frac{36}{(n-2)\theta_0(\ell)}$$

(with S' and $\theta_{S'}$ in the adaptive case), which is $36\ell_0/((n-2)\theta'_0) = O(1/\log^2 n)$ for Hypothesis 6.4, and for Hypothesis 6.7, with $m_\ell \leq n/(\log^2 n \min(\ell - 1, M)) + 1$,

$$\begin{aligned} \sum_{\ell < \ell_0} \frac{144 m_\ell}{(n-2)\kappa_2 \log^2 n} &\leq \frac{144}{\kappa_2 \log^4 n} \left(\sum_{\ell-1 \leq M} \frac{1}{\ell-1} + \sum_{M < \ell-1 < \ell_0} \frac{1}{M} \right) + \frac{144 \ell_0}{(n-2)\kappa_2 \log^2 n} \\ &= O\left(\frac{\log n}{\log^4 n} + \frac{\ell_0}{M \log^4 n} + \frac{1}{\log^4 n} \right) = O\left(\frac{1}{\log^2 n} \right). \end{aligned}$$

The exceptional events of (ii) over all ℓ form a single event of probability $o(1/\log n)$ ($\ell = d_{12}$ is a frame statistic). Altogether $\mathbb{P}[\text{Flip} = \emptyset] = o(1/\log n)$. $\square$

What the hypothesis says, and what it costs. A ladder pair enters U when one of its T candidate column pairs has a short column cycle through x_k that is clean — a configuration of

$O(\log^2 n)$ cells ($4t$ cells of rows x_k, y_k to locate the candidate, and the cycle itself). Heuristically a candidate's column cycle has length $\leq \ell_1$ with probability $\approx \ell_1/n$ (the data of §(d) show the lengths spread like a uniform variable on $\{2, \dots, n\}$), is clean with probability $\approx e^{-2s\ell_1/n}$ for a family of s pairs, and the chosen pair avoids the $\approx 2Ts$ candidate columns of the other pairs with probability $\approx (1 - 2Ts/n)^2$, so

$$\mathbb{E}|U| \approx s \cdot \frac{T\ell_1}{n} e^{-2s\ell_1/n} \left(1 - \frac{2Ts}{n}\right)^2 \asymp \log^2 n \quad \text{for } s = \frac{n}{16\log^2 n}, T = \frac{\log^2 n}{4}, \ell_1 = \frac{\log^2 n}{2},$$

against the $C \log \log n$ ($C > 1/\log 2$) that (i) needs; the constant is small ($\approx 1/145$), so the heuristic excess over the threshold is a large- n statement. Compared with the fixed matching of (c), which offers $\approx n/2$ coordinates per pair of which a constant number are short and clean and a fraction ≈ 0.19 of those separated, the adaptive rule offers only $T = O(\log^2 n)$ coordinates per pair, all separated: the genericity constant is gone, and in exchange membership in U is a rare event per pair, so (i) is a lower-tail statement about rare local events at rows selected by the square — of the same shape as a statement that some ladder pair has a short ρ_{x_k, y_k} -cycle through p avoiding p' (which would make it flippable outright), now with a column cycle of a free pair through x_k in its place. There is a second regime: with $s \approx T \approx \ell_1 \approx \sqrt{n}$ a pair enters U with probability bounded below and $\mathbb{E}|U| \asymp \sqrt{n}$, so (i) becomes a second-moment statement about $\sqrt{n}$ events of constant probability, at the price of the arcs of ρ_k at p, p' exceeding $\sqrt{n}$ for a constant fraction of the pairs. In both regimes the lengths of the arcs of ρ_k at p, p' — which the fixed-matching version needed to be macroscopic for a positive orbit weight — enter only through the number of candidates $\min(T, \mu_k - 1)$. The collision rule in U (the chosen pair must avoid the candidate columns of *all* other pairs) is what makes U orbit-invariant — a turn can shrink the candidate set of an excluded pair, so the obvious greedy refinement is not invariant — and it is crude: it is why the asymptotic regime $2Ts/n \leq \frac{1}{2}$ is out of reach of sampling at $n \leq 100$.

Comparison with the witness lemma. Hypothesis 5.1 asks for *one* witness on a long ladder (a short ρ_{x_k, y_k} -cycle through p avoiding p' , which makes the pair flippable with no further argument), at a per-pair rate $\approx \ell'/n$. Hypothesis 6.7 asks for $C \log \log n$ good candidates at a per-pair rate $\approx T\ell_1/n$, and Hypothesis 6.4 asks for a genericity constant. Neither orbit hypothesis is implied by the witness lemma or implies it (we see no implication in either direction); the witness lemma is the least demanding heuristically and is the reference hypothesis of this paper. What the orbit method contributes is the mechanism — exact fairness compounded over commuting involutions into exponential decay — and, with the adaptive coordinates, the observation that the mechanism needs only rare local witnesses of a second kind (column cycles of a free pair through x_k) rather than a constant; at the first ladder pair alone, (e) below, the rarity disappears too, and what is left is the lower tail of the number of good candidates among $\mu - 1$ column pairs. All of these hypotheses share the same unproved core, stated in §7.

(d) Data. All scripts and logs are in the repository [8], under `s13_*.py` and `runs/s13/orbit/`. *Adaptive coordinates.* On instances with a uniformly random mark at $n = 30, 50, 100$ (family the first $\min(K_0 - 1, n/8)$ ladder pairs, $T = 6$, $\ell_1 = 6, 8, 10$; 2400, 3000, 900 instances), for a random $k \in U$ the turn of Z_k was applied and checked to toggle the status of (x_k, y_k) , to leave the status, μ_l , $t_0(l)$, C_l of every other pair and the set U unchanged, and to preserve the type of rows 1, 2 and the frame (682, 697, 215 turns, no failure; 444 and 160 more with families of 12 and 30 pairs);

the bound $\mathbb{P}[\text{Flip}_I = \emptyset] \leq \mathbb{E}[2^{-|U|}]$ holds in every K_0 class. At these sizes 30%, 23%, 16% of the ladder pairs have a good candidate and 13%, 5%, 3% are in U (8% and 1.4% with the family of 30 pairs at $n = 100$): $2Ts \geq n$ there, so the collision rule removes most pairs. *Column-cycle lengths and the fixed-matching coordinates.* The length of the column cycle of a free pair through a ladder row is spread like a uniform variable on $\{2, \dots, n\}$ (about $1/n$ per length; the turn is legal for 98% of the 2-cycles and 64% of the long cycles, and avoids y for 97% resp. 27% of them, at $n = 100$). For the fixed matching of (c): $\mathbb{P}[(q, c) \text{ separated} \mid \text{clean, length } \ell] = 0.193\text{--}0.201$ for $\ell = 2, \dots, 8$ against 0.195 over all column pairs ($n = 100$), i.e. no detectable dependence on the cycle length; with $|\mathcal{T}| \approx 42$ coordinates the orbit average of the parallel bit is 0.478 from a crossed and 0.523 from a parallel start ($n = 100$); with the fixed matching of consecutive free columns $|\mathcal{T}| = 1.7$ resp. 3.0 clean coordinates per pair ($n = 50, 100$) and $\mathbb{E}[\bar{y}] = 0.159$ resp. 0.189; and the orbit weights of different ladder pairs are uncorrelated to within sampling error ($\text{Var}(Y_I)/(|I|\text{Var } \bar{y}) = 0.99$ and 1.14 at $n = 50, 100$, the excess coming from the variation of $|I|$). The ladder data of Remark 4.4 complete the picture: $\mathbb{P}[\text{Flip} = \emptyset] \approx 4/n$, as if the ladder coins were fair and independent.

(e) The first ladder pair alone. The adaptive coordinates of (c') specialise to the single pair $(x_1, y_1) = (\pi^{-1}(1), \pi^{-1}(2))$ with none of the side conditions of Proposition 6.6 (there are no other pairs to protect), and the resulting involution combines with the first ladder trade T_1 into a bound on $\mathbb{P}_X[B] - \mathbb{P}_X[A]$ by an event selected by that one pair. Since $\alpha, \beta \geq 2$ the pair always consists of rows outside $\{1, 2\}$: $x_1 = 2$ would mean $p' = \sigma(p)$ and $y_1 = 1$ would mean $p = \sigma(p')$ (so $K_0 \geq 2$ always). Write $\rho = \rho_{x_1, y_1}$, F for the event that (x_1, y_1) is flippable (parallel), and ν for the minimum of the two arc lengths from p to p' and from p' to p (crossed case) or of $|Q_p|, |Q_{p'}|$ (parallel case) — the quantity written μ_{x_1, y_1} in (c'); we use ν here because μ is the split type elsewhere in the paper.

Lemma 6.9. $\nu \geq 2$ *always*.

Proof. $\rho(p) = p'$ would mean $L(y_1, p') = L(x_1, p) = L(1, p')$, impossible in a column; $\rho(p') = p$ would mean $L(x_1, p') = L(y_1, p) = L(2, p')$, again impossible in a column; and ρ has no fixed points since $x_1 \neq y_1$. $\square$

For $1 \leq t < \nu$ the column pair $\{q_t, c_t\} = \{\rho^t(p), \rho^t(p')\}$ (written q^t, c^t in (c')) is free, consists of two distinct columns, and is separated by the mark, as observed in (c'). Let $\tau_t = \delta_{q_t, c_t}$ be the permutation of rows induced by the column pair ($\tau_t(r)$ is the row of $L(r, q_t)$ in column c_t) and $Z_t(r)$ its cycle through row r . Call t *good* if x_1 and y_1 lie in different cycles of τ_t and at least one of $Z_t(x_1), Z_t(y_1)$ meets $\{1, 2\}$ in 0 or 2 rows (so that its turn is legal); let

$$G = \{\text{some } t < \nu \text{ is good}\},$$

and on G let t_0 be the least good t and Z the admissible cycle at t_0 (the one through x_1 if it is admissible, otherwise the one through y_1).

Proposition 6.10 (single-pair involution). *Turning Z (exchanging the entries of columns q_{t_0}, c_{t_0} in the rows of Z) is an involution of $X \cap G$ that preserves the frame π (hence x_1, y_1, A and B), toggles F , and fixes t_0 and Z . Consequently $\mathbb{P}_X[F \cap G] = \mathbb{P}_X[F^c \cap G]$ and $|\mathbb{P}_X[F] - \frac{1}{2}| \leq \frac{1}{2}\mathbb{P}_X[G^c]$.*

Proof. Columns p, p' are untouched, so π, x_1, y_1 , the mark and the rows outside Z are unchanged. Rows 1, 2 are both in Z or both outside; if both are in, they become $L_1 \circ (qc)$ and $L_2 \circ (qc)$

(row 1 is then no longer the identity, which the unnormalised definition of X allows), so the new row permutation is $(qc)\sigma(qc)$, of the type of σ , and as $q, c \notin \{p, p'\}$ it still has p' at distance α from p : the turned square is in X . Exactly one of x_1, y_1 is in Z (they lie in different τ_{t_0} -cycles), so $\rho' = \rho \circ (qc)$ if $x_1 \in Z$ and $\rho' = (qc) \circ \rho$ if $y_1 \in Z$. In the first case Lemma 6.5 applies with $t = t_0$. In the second, $(qc) \circ \rho = \rho \circ (u u')$ with $u = \rho^{t_0-1}(p)$, $u' = \rho^{t_0-1}(p')$ (possibly $t_0 = 1$ and $\{u, u'\} = \{p, p'\}$), and the computation in the proof of Lemma 6.5, read with $t_0 - 1$ in place of t (for $t_0 = 1$ the paths are empty and the computation is immediate), shows that the ρ -paths from p to u and from p' to u' are unchanged, that $\rho'(u) = c$ and $\rho'(u') = q$ (the ordered pair at step t_0 is reversed, the unordered pair is $\{q, c\}$), and that crossed with arcs (a, b) becomes parallel with cycle lengths (b, a) and conversely. In both cases $\nu' = \nu$, the candidate pairs are the same for $t \leq t_0$, and F toggles. The column pairs with $t < t_0$ are disjoint from $\{q, c\}$, so their τ_t , hence their badness, are unchanged. At t_0 the turn replaces τ_{t_0} by τ_{t_0} with the cycle Z reversed (or by the inverse of that, if the ordered pair is reversed), so x_1, y_1 are still in different cycles, Z is still a cycle with the same intersection with $\{1, 2\}$, and the other cycle through $\{x_1, y_1\}$ is unchanged: t_0 and Z are the same in the turned square, and the turn is an involution. The identity follows, and $\mathbb{P}[F] \geq \frac{1}{2}\mathbb{P}[G]$, $\mathbb{P}[F^c] \geq \frac{1}{2}\mathbb{P}[G]$ give the bound. $\square$

Proposition 6.11 (the first ladder pair controls the bias). *For every $n, \lambda, \alpha, \beta$,*

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{E}_X[(\mathbf{1}_B - \mathbf{1}_A)(\mathbf{1}_{F^c} - \mathbf{1}_F); G^c];$$

consequently $|\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{1}{2}\mathbb{P}_X[G^c]$, $\mathbb{P}_X[A], \mathbb{P}_X[B] \geq \frac{1}{2}\mathbb{P}_X[G]$, and $|\bar{q}(\mu \rightarrow \lambda) - 1| \leq 2\mathbb{P}_X[G^c]/(1 - \mathbb{P}_X[G^c])$.

Proof. Write $B_{FG} = \mathbb{P}_X[B \cap F \cap G]$ and so on. The trade T_1 of Theorem 4.2 ($k = 1$) is an involution of $X \cap F$ whose frame is $(x_1 y_1) \circ \pi$, so it exchanges A and B : $B_{FG} + B_{FG^c} = A_{FG} + A_{FG^c}$. The turn of Proposition 6.10 is an involution of $X \cap G$ exchanging F and F^c and preserving π , hence A and B , which are functions of π : $B_{FG} = B_{F^cG}$ and $A_{FG} = A_{F^cG}$. Hence

$$\begin{aligned} \mathbb{P}[B] - \mathbb{P}[A] &= (B_{F^cG} + B_{F^cG^c}) - (A_{F^cG} + A_{F^cG^c}) = (B_{FG} - A_{FG}) + (B_{F^cG^c} - A_{F^cG^c}) \\ &= (A_{FG^c} - B_{FG^c}) + (B_{F^cG^c} - A_{F^cG^c}), \end{aligned}$$

which is the displayed expectation; its absolute value is at most $\mathbb{P}[G^c]$. (T_1 need not preserve G .) The last inequality is $|\bar{q} - 1| = |\mathbb{P}[B] - \mathbb{P}[A]|/\mathbb{P}[A]$ with $\mathbb{P}[A] \geq \frac{1}{2}(1 - \mathbb{P}[G^c])$. $\square$

Hypothesis 6.12 (first-pair hypothesis). $\sup_{\lambda, \alpha, \beta} \mathbb{P}_{X(n, \lambda; \alpha, \beta)}[G^c] = o(1/\log n)$.

Corollary 6.13. *Hypothesis 6.12 implies Conjecture 1.1, with $d_{TV}(P_n, Q_n) = O(\log n \cdot \sup \mathbb{P}_X[G^c] + n^{-1+o(1)})$.*

Proof. With $s = \sup \mathbb{P}_X[G^c]$, Proposition 6.11 gives $|\bar{q}(\mu \rightarrow \lambda) - 1| \leq 2s/(1 - s)$ for every adjacent pair, i.e. EQ(δ) with $\delta = 2s/(1 - s) = o(1/\log n)$, and Theorem 3.3 applies. $\square$

What the first-pair hypothesis asks, and what it costs. The event G^c is selected by the first ladder pair but is not local to it: it involves the column cycles of the pairs $\{q_t, c_t\}$ through all rows, and rows 1, 2 through admissibility. It is again a lower tail of a count of events at square-selected positions — the obstacle is unchanged in kind — and it decomposes as

$$\mathbb{P}_X[G^c] \leq \mathbb{P}_X[\nu \leq M] + \mathbb{P}_X[\nu > M, t = 1, \dots, M \text{ all bad}].$$

The first term is an upper bound on a short arc or cycle of ρ_{x_1, y_1} at p, p' ; nothing of the kind is proved at a frame-selected pair (Proposition 4.7 bounds a different frame event), and heuristically it is $\approx 4M/n$. The second term — among M candidates on disjoint free column pairs not all are bad — is the obstacle itself. Compared with Hypothesis 5.1, at the level of heuristics and data only: one row pair instead of $\geq n/\log^2 n$; per-event probability ≈ 0.4 (the random-permutation value for “ $x_1 \not\sim y_1$ under τ_t and one of the two cycles has even intersection with $\{1, 2\}$ ”) instead of ℓ'/n ; expected count $\approx 0.4 \mathbb{E}\nu$ with $\mathbb{E}\nu = 6.3, 9.6, 18$ at $n = 30, 50, 100$ in the data (Remark 6.14), instead of $\geq \log^2 n$ ($\log^3 n$ at the parameters of Proposition 5.2); no offset families, no splice bound, no conditioning on K_0 . Neither hypothesis is known to imply the other: $\mathbb{P}_X[F^c] \geq \frac{1}{2} \mathbb{P}_X[G]$, so the first pair alone cannot give (L), and Hypothesis 5.1 says nothing about G . So Hypothesis 6.12 is a third alternative sufficient condition, the one with the fewest moving parts; the reference hypothesis of the paper remains Hypothesis 5.1. The minimal instance (6) of §7 follows from Hypothesis 6.12 (with $c = \frac{1}{2} - o(1)$) but not evidently from Hypothesis 5.1, and is sufficient for neither.

Remark 6.14 (data: completions of a fixed rectangle). To test the hypotheses on types that uniform sampling never produces we ran the Jacobson–Matthews chain restricted to the completions of a fixed $2 \times n$ rectangle (`jmfix.c`: moves touching rows 1, 2 are never proposed from a proper state and are rejected from an improper one; the walk is a simple random walk on the restricted move graph, every proper state having restricted degree $(n-2)n(n-3)$, so its stationary law is uniform on the proper states of the component of the start). Connectivity of the restricted graph is not proved; at $n = 5, 6$ (types (5), (6), (2, 2, 2); 36, 4032, 5376 completions, enumerated exactly) every completion is reached from two different starts with χ^2 statistics within one standard deviation of uniformity, and at $n = 7$ the sampled $\mathbb{P}[B \mid R] = 0.4965$ agrees with the exact 0.497076 over all 6566400 completions of $R = (\text{id}, (1\,2 \cdots 7))$. Mixing is not analysed; runs with different thinning and starts agree within noise. Any two rectangles of type λ are isotopic by a column permutation and a symbol relabelling fixing rows 1, 2, which preserve completion counts and carry a mark of class (m, α) to one of the same class; every event considered here is defined through the cycle structures of row and column permutations and is invariant under such isotopies. Hence $\mathbb{P}_X$ of such an event equals its probability under a uniform completion of any one rectangle R of type λ with any one mark of the class, and averaging over the marks of the class in R (which are equivalent under the isotopies preserving R) only reduces the variance. Findings (assuming connectivity and mixing of the sampler) ($n = 30$: $6 \cdot 10^4$ squares per type; $n = 50$: $4 \cdot 10^4$; $n = 100$: $1.2 \cdot 10^4$; types (n) , $(n/2, n/2)$, $(n-2, 2)$, $(10, 10, 10)$, $(4, 2^{13})$, $(6, 3^8)$, $(4, 2^{23})$, $(6, 3^{14}, 2)$, $(4, 2^{48})$; marks at $\alpha = 2$, $\approx m/3$, $\lfloor m/2 \rfloor$): (i) $\mathbb{P}_X[B] \in [0.4974, 0.5024]$ in every class at $n \leq 50$ (sampling error ≈ 0.0025), i.e. by Theorem 2.4 $C_n(\lambda)/C_n(\mu) \in [0.995, 1.005]$ for every tested adjacent pair, including $(4, 2^{13}) \rightarrow (2^{15})$ and $(6, 3^8) \rightarrow (3^{10})$, types of P_n -probability $e^{-\Theta(n)}$; $[0.495, 0.504]$ at $n = 100$ (error ≈ 0.005). (ii) $\mathbb{P}_X[F] \in [0.4988, 0.5017]$ in every class at $n \leq 50$ (0.4896 at $n = 7$, so $\frac{1}{2}$ is not an identity). (iii) Along the ladder, $\mathbb{P}[\text{no witness among the first } K \text{ pairs} \mid K_0 - 1 \geq K]$ equals $(1 - w_1)^K$ within sampling error for $K \leq 12, 20, 40$ at $n = 30, 50, 100$ and thresholds $\ell' = 4, 8, 16, 24$, where w_1 is the first-pair witness rate (0.100, 0.060, 0.030 at $\ell' = 4$); the hazard $\mathbb{P}[\text{witness at } k \mid \text{none before } k]$ is flat in k to three digits; $\mathbb{P}[\text{none of the first } K \text{ ladder pairs flippable} \mid K_0 - 1 \geq K] = 2^{-K}$ within noise to $K = 12$ – 15 ($7 \cdot 10^5$ instances at $n = 30$); across the diagonals D_c of B -instances with $d_{12} = \ell \leq n/4$, which share their y -rows, the number of diagonals carrying a witness vanishes with the binomial probability and has binomial lower tails, for all $\ell \leq 12$. All of this is identical for the extreme types. (iv) For Hypothesis 6.12, with all candidates $t < \nu$: $\mathbb{P}_X[G^c] = 0.61, 0.18, 0.11, 0.055$ at

$n = 7, 30, 50, 100$ (type (n) , $\alpha = 2$; 0.056 for $\alpha = 50$ at $n = 100$), i.e. $\approx 5.5/n$ for $n \geq 30$; $\mathbb{E}\nu = 6.3, 9.6, 18$ and $\mathbb{P}[\nu = k] \approx 4/n$ for small k ; $\mathbb{P}[G^c \mid \nu = k] = 0.59, 0.37, 0.22, 0.13, 0.08$ for $k = 2, \dots, 6$ at $n = 30$ against 0.59^{k-1} (a slight positive dependence among the candidates, which share the rows x_1, y_1); $t = 1$ is good with probability 0.41 in every class at $n \leq 100$. Other types were measured only with $t \leq 8$: $\mathbb{P}[\text{some } t \leq 8 \text{ good}] = 0.81, 0.88, 0.94$ at $n = 30, 50, 100$ in every class. The identity of Proposition 6.11 and its two ingredients hold to all digits on the exact enumerations at $n = 5, 6$ (every class) and within sampling error at $n = 7$ (-0.0052 and -0.0058 , each ± 0.001 , against the exact -0.005848). None of these data reach the regime of any hypothesis ($\ell' \geq \log^5 n$, failure $o(1/\log n)$); they exclude a visible dependence between the events, or a visible effect of the type, at the first- and second-moment level for $n \leq 100$.

7 What remains

Hypotheses 5.1, 6.4, 6.7 and 6.12 are statements about configurations at positions selected by the frame: a short ρ_{x_k, y_k} -cycle through p avoiding p' ; a clean column pair separated by the mark; a short clean column cycle at a prescribed column pair; a column pair along ρ_{x_1, y_1} whose column permutation separates x_1 from y_1 . Their heuristic frequencies are, for large n , far above what is needed; the difficulty is entirely that the positions are selected by the frame of the mark and the whole space is conditioned on the type of rows 1, 2. The minimal instance of the obstacle is a lower bound on a single frame-selected pair:

$$\mathbb{P}_X[(x_1, y_1) \text{ flippable}] \geq c, \quad x_1 = \pi^{-1}(1), \quad y_1 = \pi^{-1}(2), \quad (6)$$

which no identity of this paper gives unconditionally: flippability of a ladder pair is a class invariant of the ladder trades, and Proposition 6.10 reduces (6) one way to $\mathbb{P}_X[G] \geq 2c$, a lower bound for an event of the same kind with rows and columns exchanged (the column permutation of a pair of columns selected from rows x_1, y_1 separating the frame-selected rows x_1, y_1), which is not known to be easier. By row exchangeability (6) is a statement at fixed positions under conditioning: it is the probability that p, p' lie in different cycles of $\rho_{3,4}$ given rows 1, 2 and the two cells $L(3, p) = L(1, p')$, $L(4, p) = L(2, p')$. Even without the conditioning on rows 1, 2 we know of no lower bound in the literature for the related quantity $\mathbb{P}[\text{the } \rho_{3,4}\text{-cycle through } p \text{ has length } k \text{ and avoids } p'], 3 \leq k \leq \log^5 n$; the one case we know to be settled is $k = 2$, where the concentration of the number of intercalates at $n^2/4$ [7] and the exchangeability of row pairs and of columns give $\approx 1/n$. We record what the available tools do and do not give.

Unconditional facts. For two generic rows of a uniform square, Proposition 3.2(i) gives, by Markov's inequality on the number of columns on short cycles, $\mathbb{P}[\text{all cycles of } \rho_{x,y} \text{ have length } \leq \varepsilon n] \leq 2\varepsilon/(1 - \varepsilon)$; so a generic pair has a cycle longer than $n/5$ with probability $\geq \frac{1}{2}$, and the column-cycle analogue holds by transposition. Kwan and Sudakov's bound $\mathbb{P}[N_2 \geq t] = e^{-\Omega(t \log t)}$ on the number of 2-cycles of two fixed rows [6, Thm. 3] is proved by joining switchings of the same kind as those of Proposition 3.2 and is likewise an upper bound on short cycles. The number of completions of any $2 \times n$ rectangle is within $e^{O(n \log^2 n)}$ of any other (van der Waerden and Brégman; [6, Prop. 5]), so any property of uniform squares with failure probability $e^{-\omega(n \log^2 n)}$ — such as the intercalate concentration of Kwan, Sah and Sawhney [7] — holds conditionally on any fixed rows 1, 2 and mark, hence in X for every λ, α, β ; but the natural failure probability of a statement about $m \leq n$ prescribed row pairs is $e^{-\Omega(m)}$, which this transfer does not reach.

Why the fixed-matching version is harder. With a fixed matching the orbit weight $\bar{y}_k$ of Proposition 6.3 needs a lower bound on the probability that a generic column pair is interleaved with $\{p, p'\}$ in ρ_{x_k, y_k} — an instance of the statement “two given pairs of lines of a random Latin square are interleaved in the cycle structure of a third pair with probability in $[c, 1 - c]$ ” inside X . Without the conditioning its upper half is elementary and its lower half reduces to macroscopic cycles (by conjugation invariance the interleaving probability given the cycle type of $\rho_{x, y}$ is $\frac{1}{3} \sum_i (m_i)_4 / (n)_4 + 2 \sum_{i \neq j} (m_i)_2 (m_j)_2 / (n)_4 \leq \frac{1}{3} + O(1/n)$, and $\geq \frac{1}{3} \mathbb{P}[W \mid \text{type}]^3 - O(1/n)$ by Jensen, W the event that two given columns share a cycle). Inside X the exact tools are the free trades of rows other than 1, 2 (which preserve interleaving) and the legal column turns (which are legal exactly off the interleaved configurations), and every identity they produce is fair: it equates two expectations with random weights whose lower bounds are again of the same form. Lower bounds on “different cycles” events are, in the language of [1], of the type of the direction of their Lemma 3.12 whose proof has the gap: they require a repair of the same-cycle configurations with a rigid inverse, and the repairs available in X (trades or cross-switches of a free pair of third rows) have unbounded multiplicity. The one case in which we found a rigid inverse is CGW’s own case 3 with $\min(\alpha, \beta) = 2$, which gives $\mathbb{P}_X[A] \geq \frac{1}{4}$ for marks at distance 2 (Remark 3.4); it does not extend to other distances for a reason of type rather than multiplicity. The adaptive coordinates sidestep the question: the coordinate is placed where it is separated by construction, and the price is paid in rarity rather than in a constant we cannot prove; at the first ladder pair alone (§6(e)) the rarity disappears as well, and what is left is an upper bound on a short arc at the frame-selected pair together with the lower tail of the number of good candidates among the $\nu - 1$ column pairs along ρ_{x_1, y_1} — the smallest sufficient condition we have, and still a lower bound at square-selected positions.

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